

Lecture 8

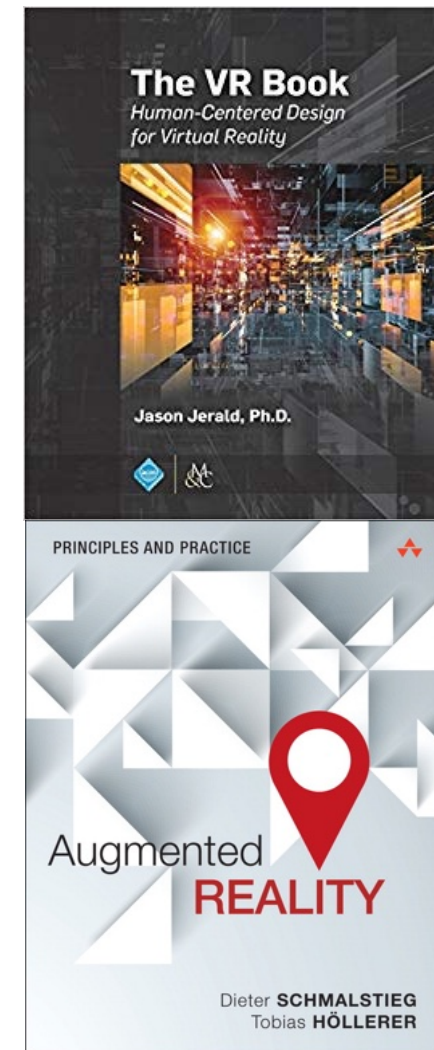
Mixed Reality

Presence, Interaction, Health



References

- Jason Jerald; **The VR Book: Human-Centered Design for Virtual Reality**, ACM, 2016.
- Dieter Schmalstieg, Tobias Höllerer; **Augmented Reality**, Addison-Wesley, 2016.
- Mark Billinghurst, <https://www.slideshare.net/marknb00/>



Mixed Reality

- Mixed Reality Concepts
- Virtual Reality
- Augmented Reality
- History
- **Immersion, Presence and Perception**
- **MR Technology**
- **VR Interaction**
- **Health Issues**

Immersion, Presence and Perception

Presence

*“The **subjective** experience of being in one place or environment even when physically situated in another”*



Witmer, B. G., & Singer, M. J. (1998). Measuring presence in virtual environments: A presence questionnaire. *Presence: Teleoperators and virtual environments*, 7(3), 225-240.

Immersion vs. Presence

Immersion: the extent to which technology delivers a vivid illusion of reality to the senses of a human participant.

Presence: a state of consciousness, the (psychological) sense of being in the virtual environment.

- **Immersion** produces a sensation of **Presence**
- **Goal of VR** is to create a high degree of Presence, making people believe they are **really** in Virtual Environment

Slater, M., & Wilbur, S. (1997). A framework for immersive virtual environments (FIVE): Speculations on the role of presence in virtual environments. *Presence: Teleoperators and virtual environments*, 6(6), 603-616.

How to Create Strong Presence?

Use Multiple Dimensions of Presence:

- Create rich multi-sensory VR experiences
- Include social actors/agents that interact with user
- Have environment respond to user

What Influences Presence?

- Vividness – ability to provide rich experience (Steuer 1992)
- Using Virtual Body – user can see themselves (Slater 1993)
- Internal factors – individual user differences (Sadowski 2002)
- Interactivity – how much users can interact (Steuer 1992)
- Sensory, Realism factors (Witmer 1998)

Example: UNC Pit Room

Key Features

- Training room and pit room
- Physical walking
- Fast, accurate, room scale tracking
- Haptic feedback – feel edge of pit, walls
- Strong visual and 3D audio cues

Task

- Carry object across pit
- Walk across or walk around
- Dropping virtual balls at targets in pit



http://wwwx.cs.unc.edu/Research/eve/walk_exp/

Typical Subject Behavior



Note – from another pit experiment
<https://www.youtube.com/watch?v=VVAO0DkoD-8>

Benefits of High Presence

Leads to greater engagement, excitement and satisfaction

- Increased reaction to actions in VR

People more likely to behave like in the real world

- E.g. people scared of heights in real world will be scared in VR

More natural communication (Social Presence)

- Use same cues as face to face conversation

Note: The relationship between Presence and Performance is unclear – still an active area of research

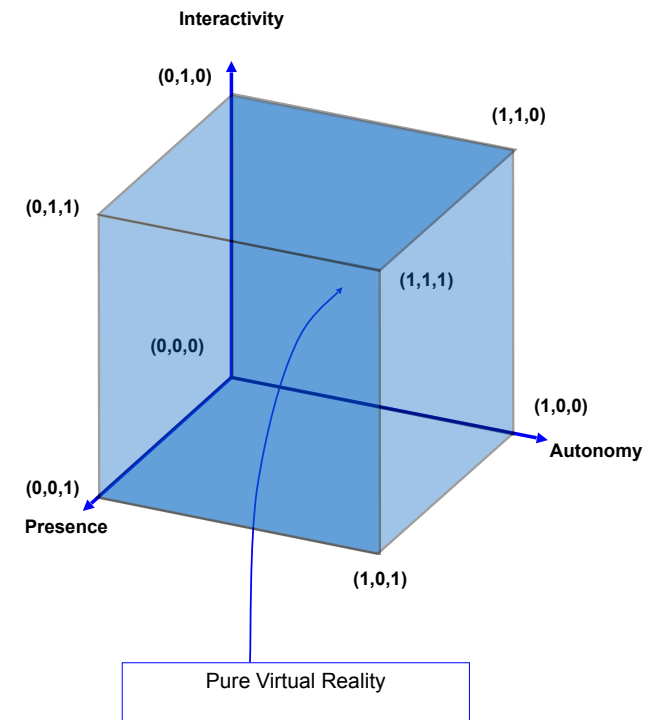


Virtual Presence

Virtual Environments Characterization

- **Zeltzer Model**

- Autonomy
 - Reaction to virtual objects, events and stimulus.
 - 0 = No reaction; 1 = complete reaction
- Interaction
 - Degree of access to parameters or variables of an object
 - 0 = off-line control
- Presence
 - Gross measure of fidelity of the input and output sensorial channels.



Virtual Presence

- Determinants of presence:
 - Dimensions of sensory information (vision, hearing, touch, ...)
 - Ability of the observer to modify the viewing position
 - Visual parallax
 - Binaural hearing
 - Ability to alter spatial relationships between objects (agency)
 - Closed loop/feedback
 - Ex: realism of the dynamic behavior of objects

Measuring Presence

Questionnaires

- IPQ presence questionnaire
<https://www.igroup.org/pq/ipq/download.php>
- Witmer & Singer 1998
<https://cs.uky.edu/~sgware/reading/papers/witmer1998measuring.pdf>
- Portuguese version: Silva et al. 2016
[A questionnaire for measuring presence in virtual environments: factor analysis of the presence questionnaire and adaptation into Brazilian Portuguese](#)

How do we perceive reality?

We understand the world through our senses:

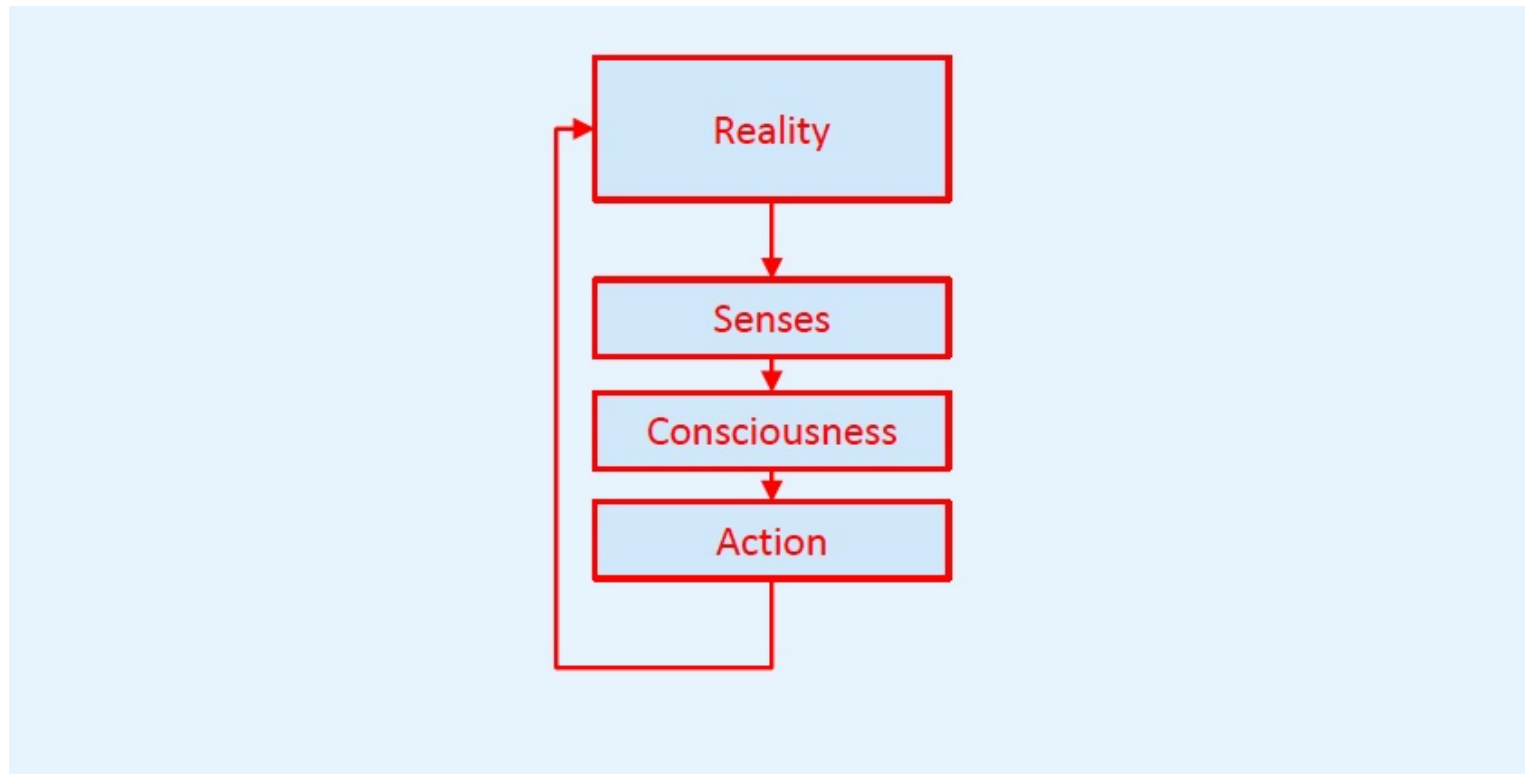
- Sight, Hearing, Touch, Taste, Smell (and others..)

Two basic processes:

- **Sensation** – Gathering information
- **Perception** – Interpreting information



Simple Sensing/Perception Model

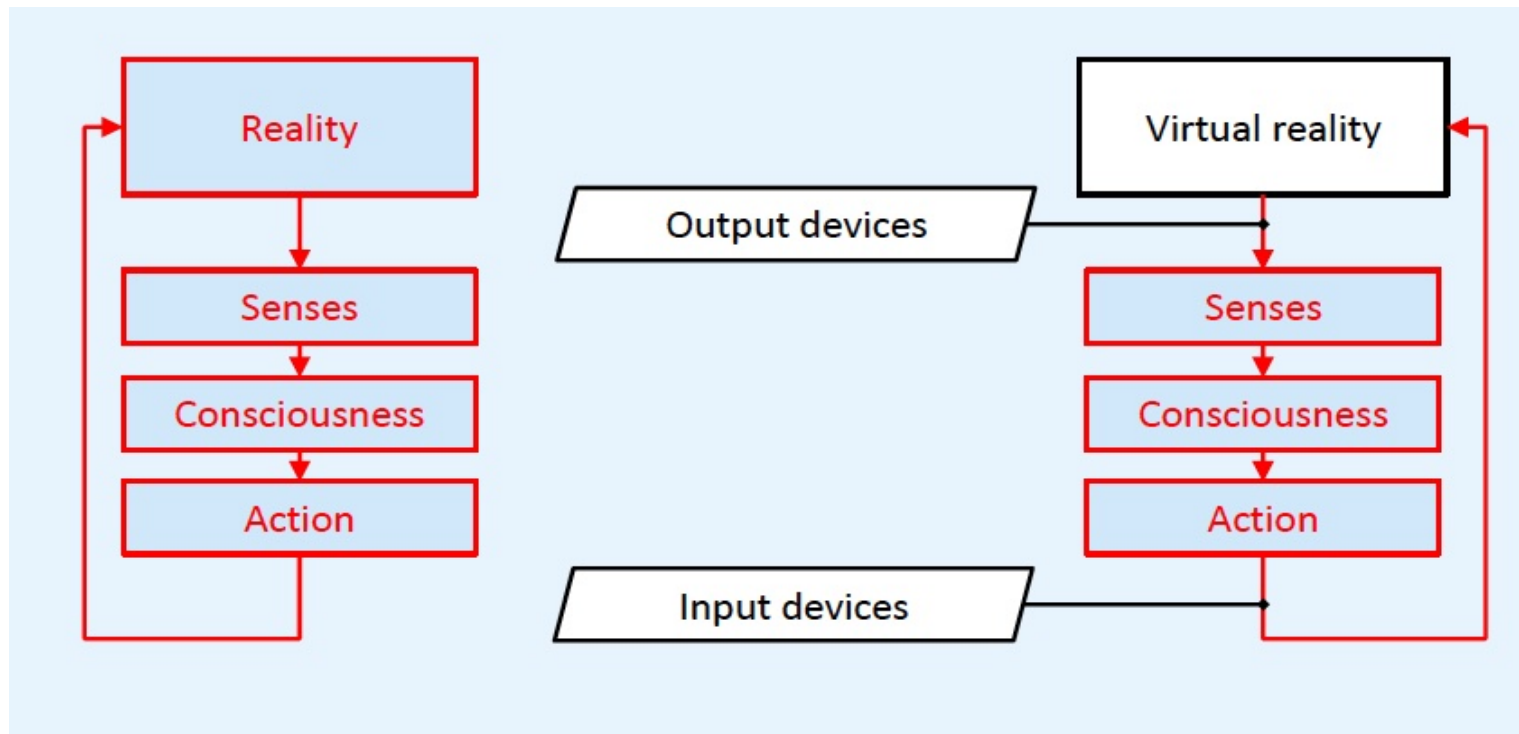


Creating the Illusion of Reality

- Fooling human perception by using technology to generate artificial sensations
- Computer generated sights, sounds, smell, etc



Reality vs. Virtual Reality



In a VR system there are **input** and **output devices** between human perception and action

Birdly Demo

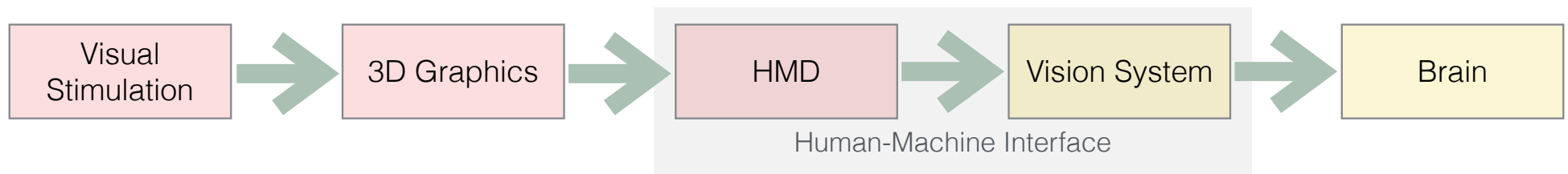
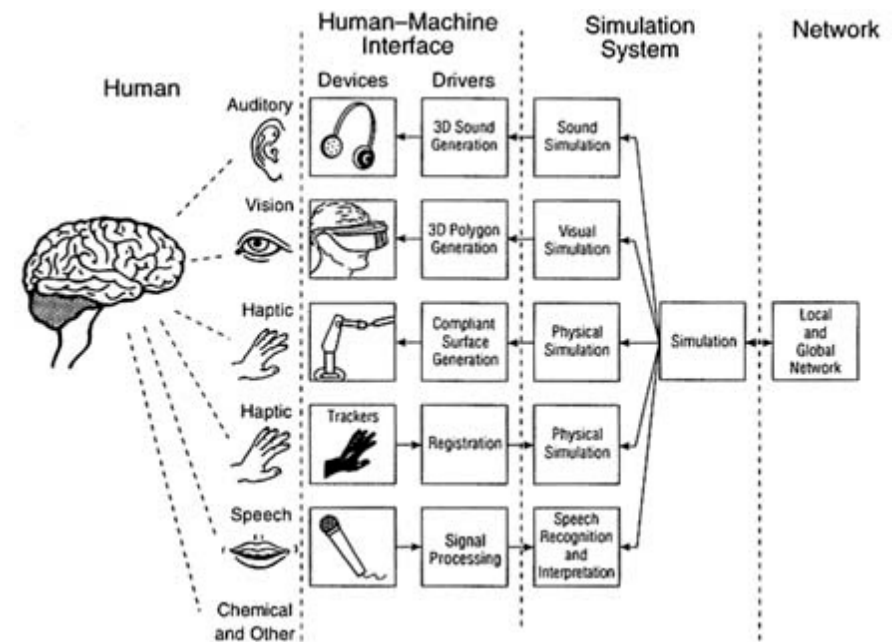


<https://www.youtube.com/watch?v=gHE6H62GHoM>

MR Technology

Using Technology to Stimulate Senses

- Simulate output
 - E.g. simulate real scene
- Map output to devices
 - Graphics to HMD
- Use devices to stimulate the senses
 - HMD stimulates eyes
- Example: Visual Simulation



Key Technologies for VR Systems

- Visual Display
 - Stimulate visual sense
- Audio/Tactile Display
 - Stimulate hearing/touch
- Tracking
 - Changing viewpoint
 - User input
- Input Devices
 - Supporting user interaction



Visual Display

Head Mounted Display

- Immerse the eyes

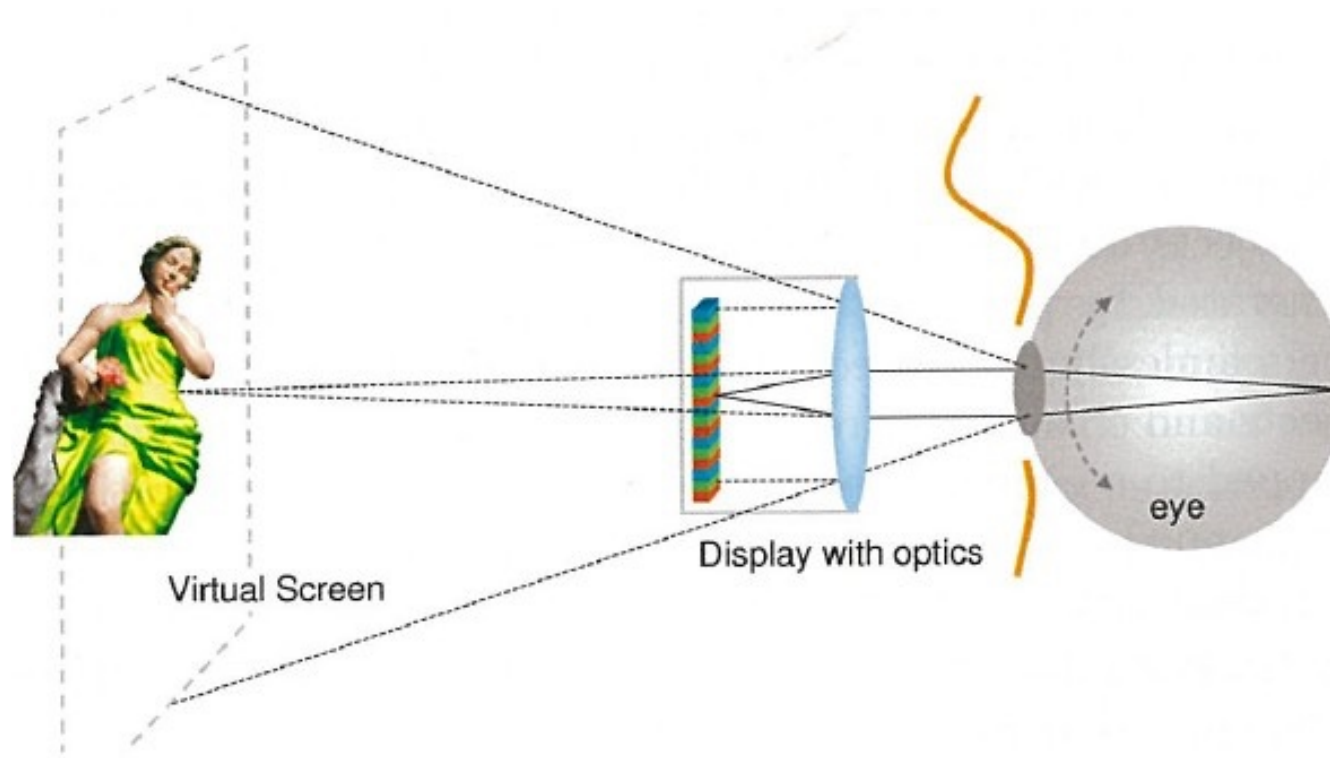
Projection/Large Screen

- Immerse the head/body

Future Technologies

- Neural implants
- Contact lens displays, etc

HMD Basic Principles



Use display with optics to create illusion of virtual screen

Key Properties of HMDs

Lens

- Focal length, Field of View
- Occularity, Inter-pupillary distance
- Eye relief, Eye box

Display

- Resolution, contrast
- Power, brightness
- Refresh rate

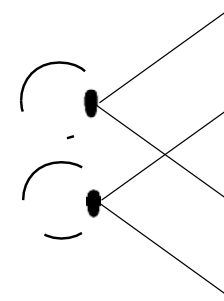
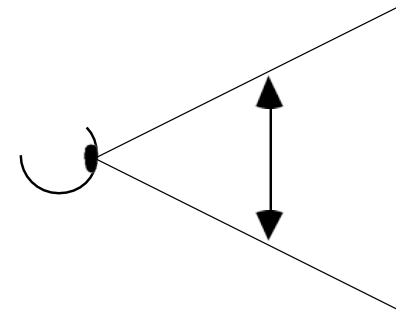
Ergonomics

- Size, weight
- Wearability



Field of View

- **Monocular FOV** is the angular subtense (usually expressed in degrees) of the displayed image as measured from the pupil of one eye.
- **Total FOV** is the total angular size of the displayed image visible to both eyes.
- **Binocular(or stereoscopic) FOV** refers to the part of the displayed image visible to both eyes.
- **FOV** may be measured horizontally, vertically or diagonally.



Ocularity



- **Monocular** - HMD image to only one eye.
- **Biocular** - Same HMD image to both eyes.
- **Binocular** (stereoscopic) - Different but matched images to each eye.



Interpupillary Distance (IPD)

- IPD is the horizontal distance between a user's eyes.
- IPD is the distance between the two optical axes in a binocular view system.

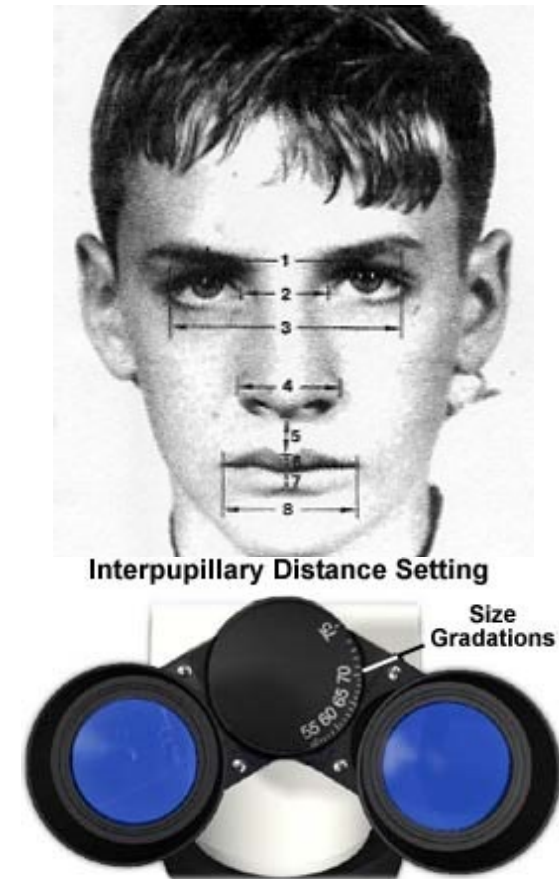
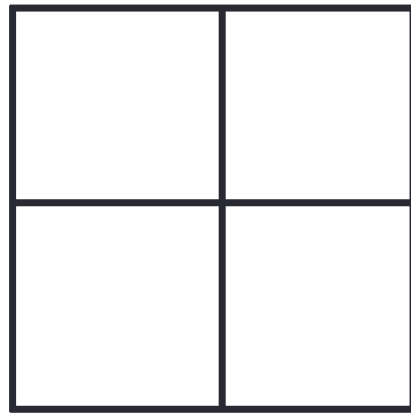
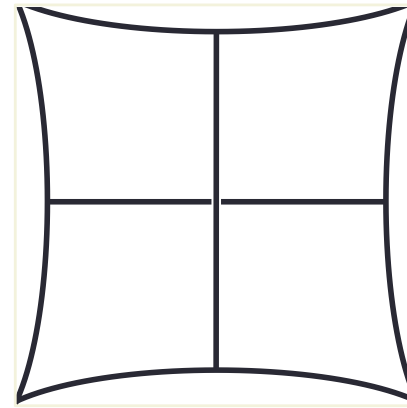


Figure 5

Distortion in Lens Optics



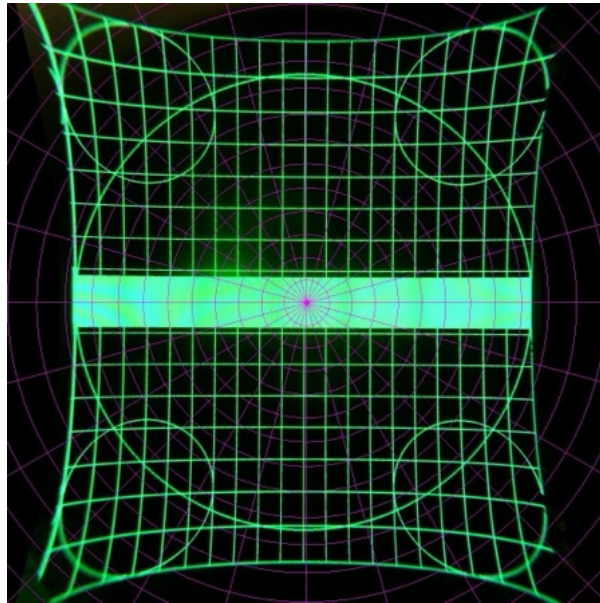
A rectangle



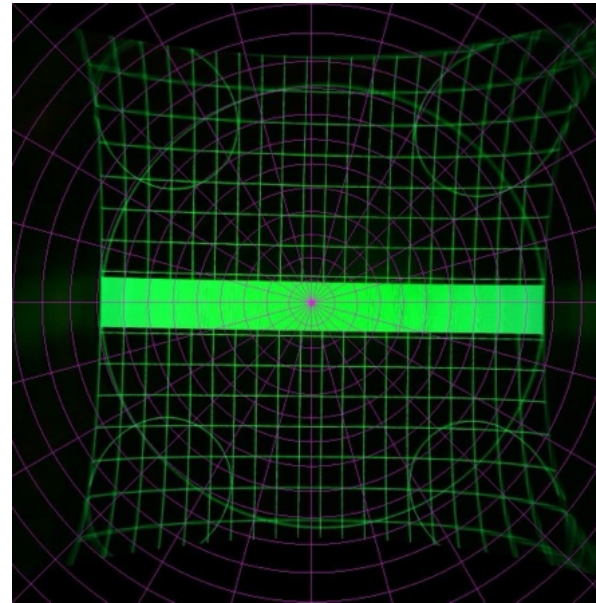
Maps to this

HMD optics distort images shown in them

Distortion Example



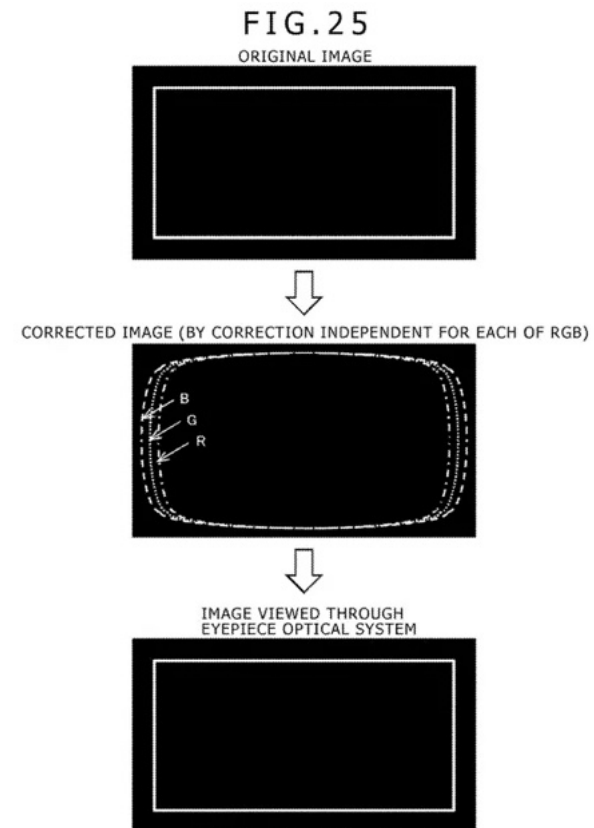
Oculus Rift DK2



HTC Vive

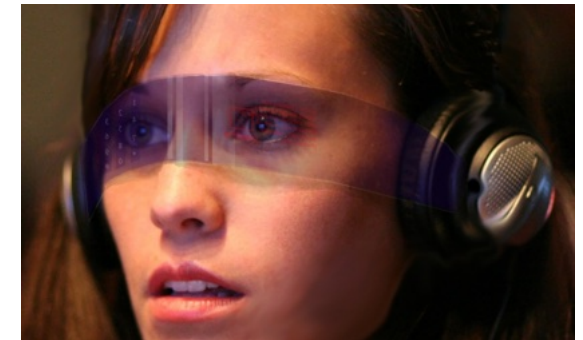
Distortion Compensation

- Simpler/Cheaper optics create distortion and **chromatic** aberration
- Images generated need to be pre-processed to create the inverse effect so that optical distortion presents the original image
- Use GPU shaders to pre-process the original images



HMD Design Trade-offs

- Resolution=Pixels Per Degree (PPD) vs. Field Of View (FOV)
 - Increase FOV $\Rightarrow$ Decrease PPD
- Eye Box (3D region eye can be) vs. FOV
 - Greater FOV $\Rightarrow$ Smaller Eye Box
 - Larger Eye Box $\Rightarrow$ Lower FOV
- Size, Weight and Power vs. everything else



Oculus Rift

- FOV: 110° Horizontal
- Refresh rate: 90 Hz
- Resolution 1080x1200/eye
- 3 DOF orientation tracking
- 3 axis positional tracking



Inside Oculus Rift DevKit2



Oculus DK2 Teardown

- Samsung 5.7" AMOLED: 1920x1080px, 75Hz
- 2 sets of lenses (for different prescriptions)
- InvenSense 6-axis IMU
- ARM Cortex-M3 MCU
- ...



<https://www.ifixit.com/Teardown/Oculus+Rift+Development+Kit+2+Teardown/27613>

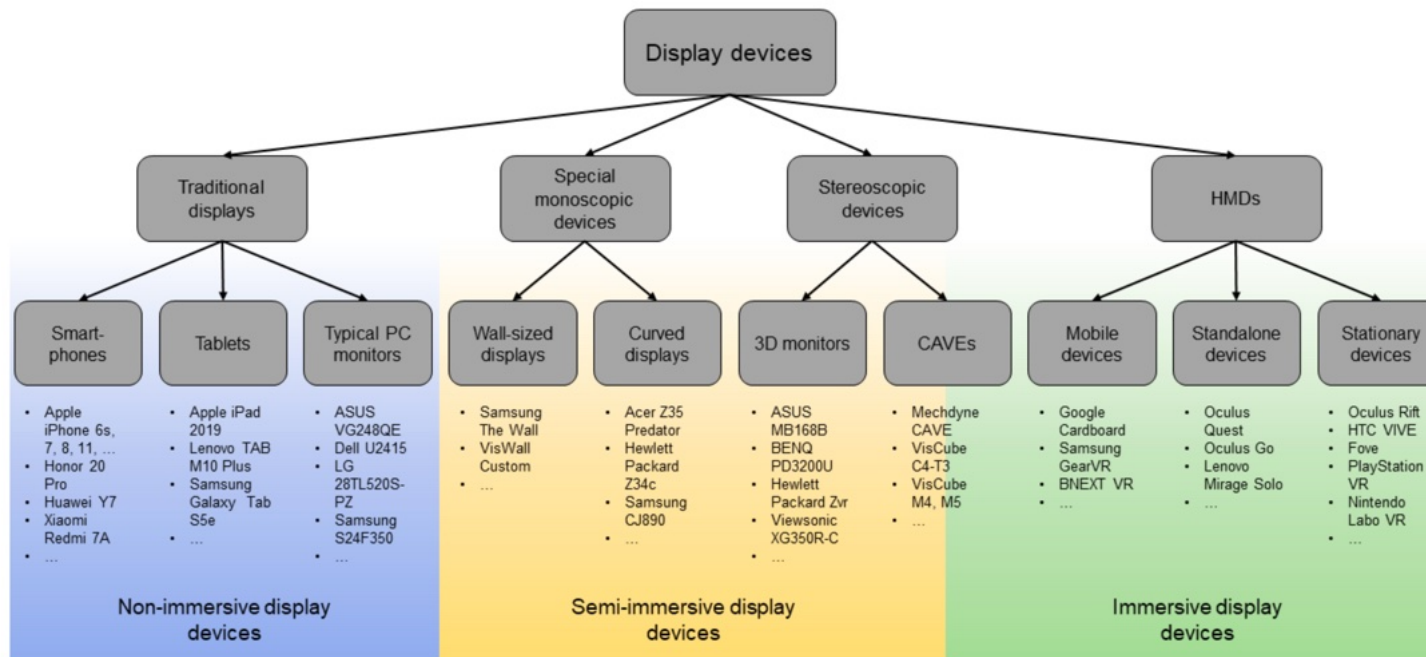
<https://youtu.be/-HoHkFgslJc>

Comparison Between HMDs (2015-17)

					
Name	Oculus Rift	HTC Vive	PlayStation VR	StarVR	OSVR HDK
Manufacturer	Oculus VR	HTC, Valve	Sony	Starbreeze	Razer, Sensics
Display	2x OLED	2x OLED	OLED	2x LCD	LCD
Resolution	2160x1200px	2160x1200px	1920x1080px	5120x1440px	1920x1080px
Framerate	90fps	90fps	120fps	60fps	60fps
Field of view	>110°	>110°	100°	210°	100°
Positional tracking	6DOF	6DOF Valve Lighthouse	6DOF	6DOF	6DOF
Controller	Xbox One controller/Oculus Touch	two SteamVR controllers, one for each hand	Playstation Move/DualShock 4	-	-

	Meta Quest 3	Apple Vision Pro	Valve Index	Microsoft HoloLens 2	Magic Leap 2
					
Resolution (per eye)	2064×2208	3660×3200	1440×1600	1440×936	1440×1760
Refresh Rate	120 Hz	100 Hz	144 Hz	60 Hz	Ø
Field of View (FOV)	~110°	~100°	~108°	~43°	~44°
Positional Tracking	6-DOF	6-DOF	6-DOF	6-DOF	6-DOF
Tracking Method	Inside-out	Inside-out	Base stations PC-powered VR (tethered)	Inside-out	Inside-out
Device Type	Standalone VR	Standalone VR	Base stations PC-powered VR (tethered)	Standalone AR	Standalone AR
Supported Reality	AR & VR	AR & VR	VR only	AR only	AR only
Release Year	2023	2024	2019	2019	2022

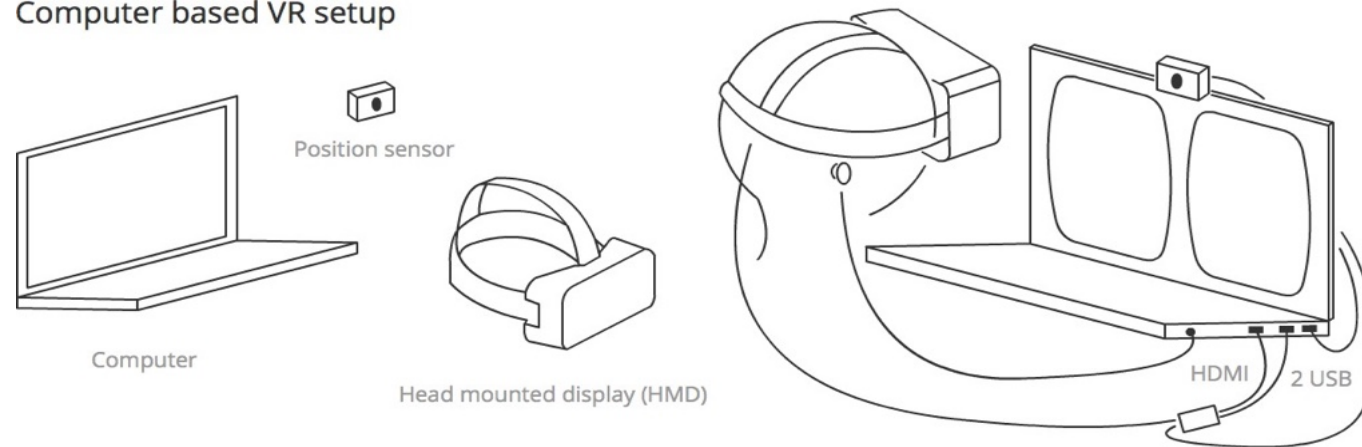
Types of MR Devices



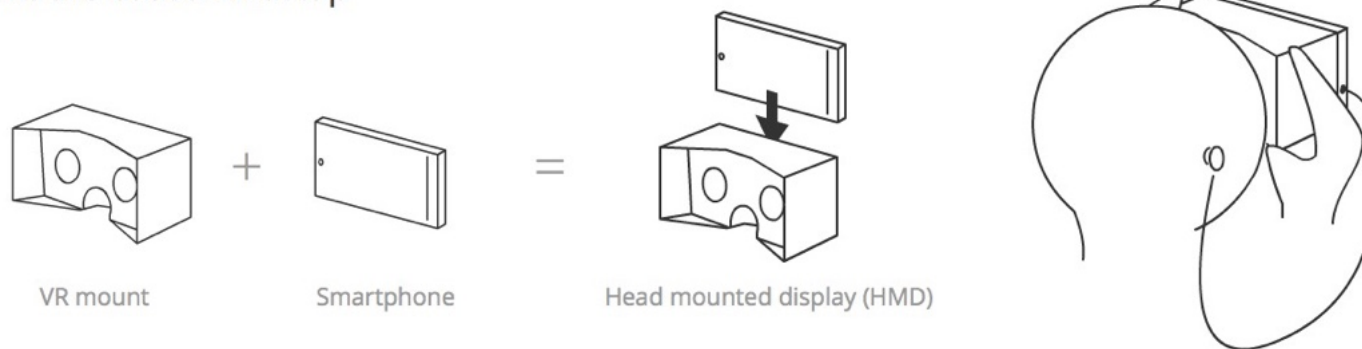
Lochhead, Ian & Madden, Marguerite & Christophe, Sidonie & Devaux, Alexandre & Pettit, Christopher & Lock, Oliver & Shukla, Shashwat & Herman, Lukáš & Stachoň, Zdeněk & Kubicek, Petr & Snopková, Dajana & Bernardes, Sergio & Hedley, Nicholas. (2020). Extended Reality in Spatial Sciences: A Review of Research Challenges and Future Directions. *International Journal of Geo-Information*. 9. 10.3390/ijgi9070439.

Computer Based vs. Mobile VR Displays

Computer based VR setup



Mobile based VR setup



Projection/Large Display Technologies

Room Scale Projection

- CAVE, multi-wall environment



<https://www.youtube.com/watch?v=rCJ1UPbWBPo&t=17s>

Dome projection

- Hemisphere/spherical display
- Head/body inside



<https://cac.es/noticia/balance-lasnocturnas-hemisferic/>

Vehicle Simulator

- Simulated visual display in windows or projections

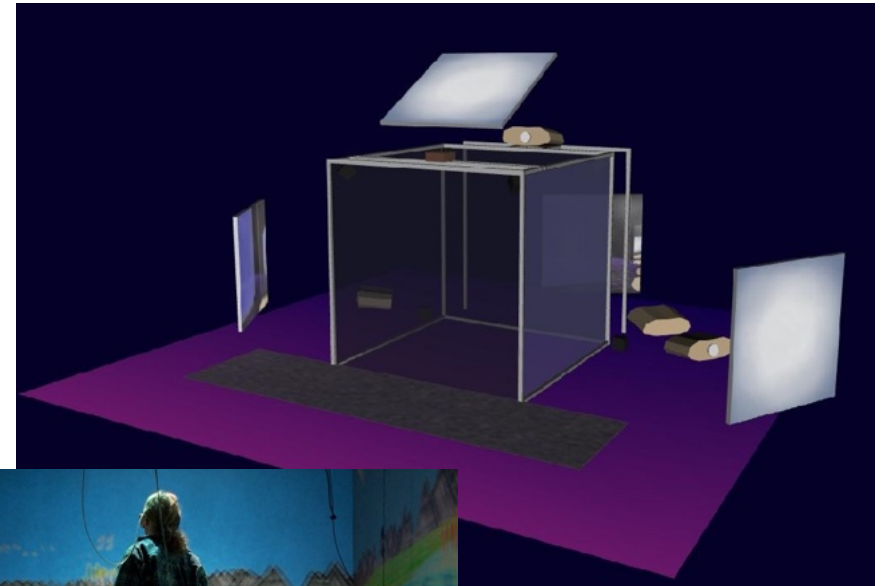


<https://flightsimbuilder.com/>

CAVE

- Developed in 1992, EVL University of Illinois Chicago
- Multi-walled stereo projection environment
- Head tracked active stereo

Cruz-Neira, C., Sandin, D. J., DeFanti, T. A., Kenyon, R. V., & Hart, J. C. (1992). The CAVE: audio visual experience automatic virtual environment. Communications of the ACM, 35(6), 64-73.



Stereo Projection

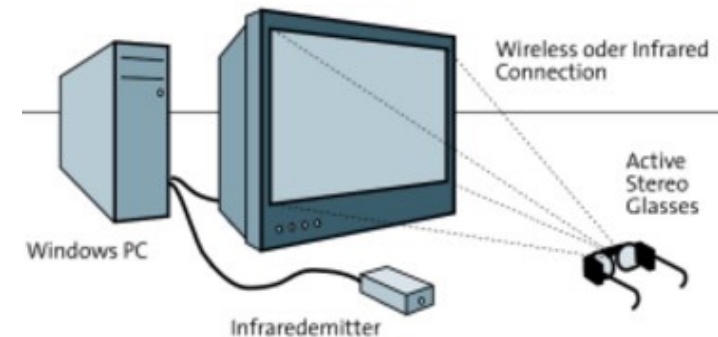
Active Stereo

- Active shutter glasses
- Time synced signal
- Brighter images
- More expensive

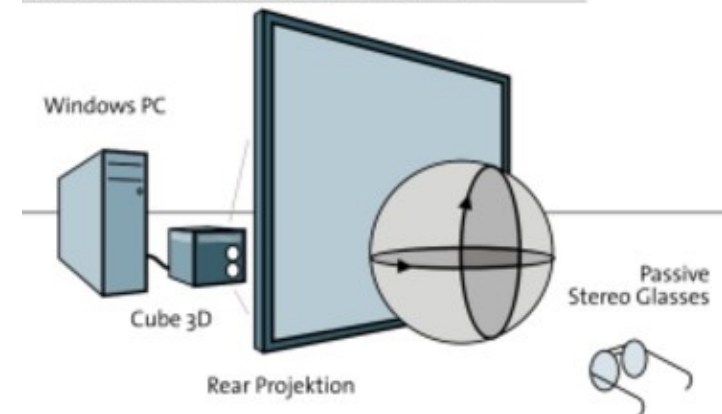
Passive Stereo

- Polarized images
- One vs Two projectors (per eye)
- Cheap glasses (powerless)
- Lower resolution (one projector)
- Dimmer
- Less expensive

ACTIVE STEREO PC SET-UP



PASSIVE STEREO PROJECTION SET-UP



Vehicle Simulators

Combine VR displays with vehicle

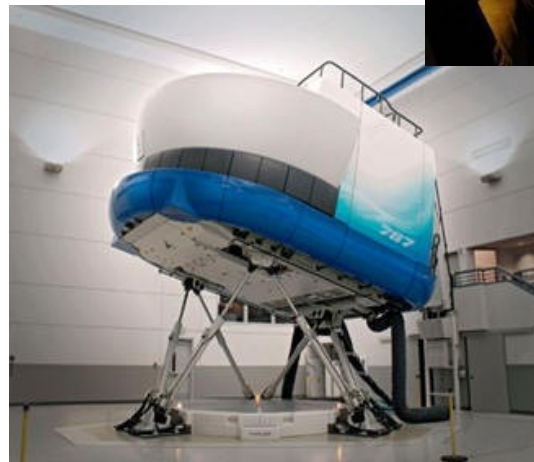
- Visual displays on windows
- Motion base for haptic feedback
- Audio feedback

Actual Physical vehicle controls

- Steering wheel, flight stick, etc

Full vehicle simulation

- Emergencies, normal operation, etc
- Weapon operation
- Training scenarios



Demo: Boeing 787 Simulator



<https://www.youtube.com/watch?v=lulF2NnBqXM>

Haptic Feedback

- Greatly improves realism
- Hands and wrist are most important
 - High density of touch receptors
- Two kinds of feedback:

Touch Feedback

- information on texture, temperature, etc.
- Does not resist user contact

Force Feedback

- information on weight, and inertia.
- Actively resists contact motion

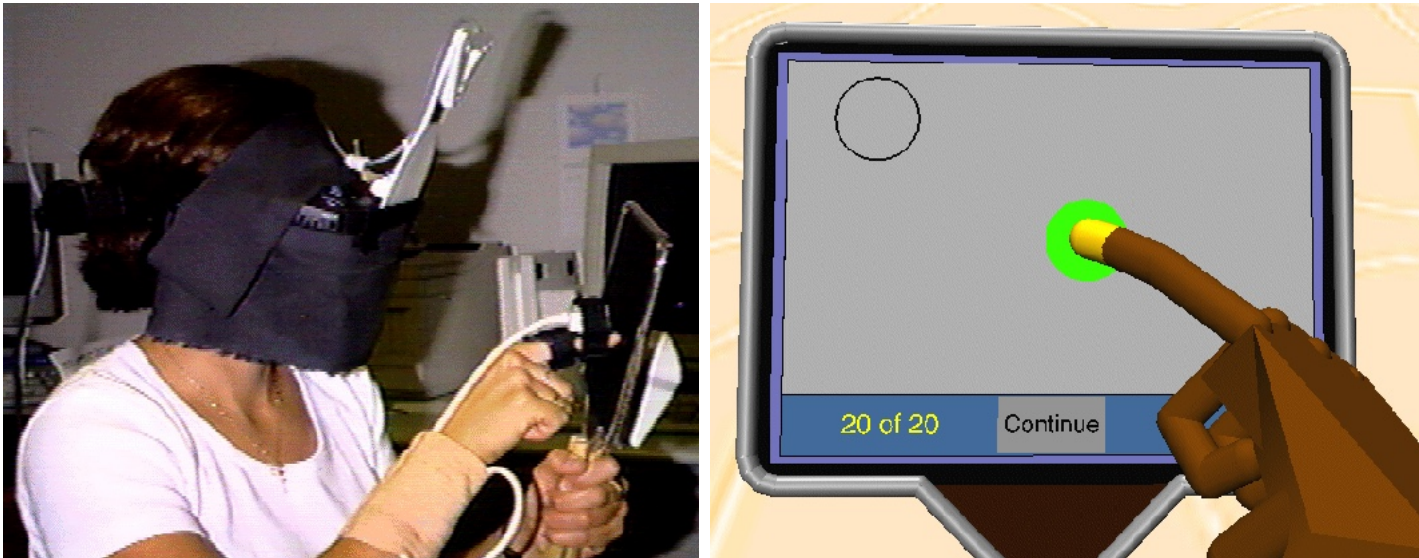


Passive Haptics

- **Not controlled by system**
 - Use real props
- **Pros**
 - Cheap
 - Large scale
 - Accurate
- **Cons**
 - Not dynamic
 - Limited use



Passive Haptic Paddle



Using physical props to provide haptic feedback
<http://www.cs.wpi.edu/~gogo/hive/> (DEAD LINK)

Tactile Feedback Interfaces

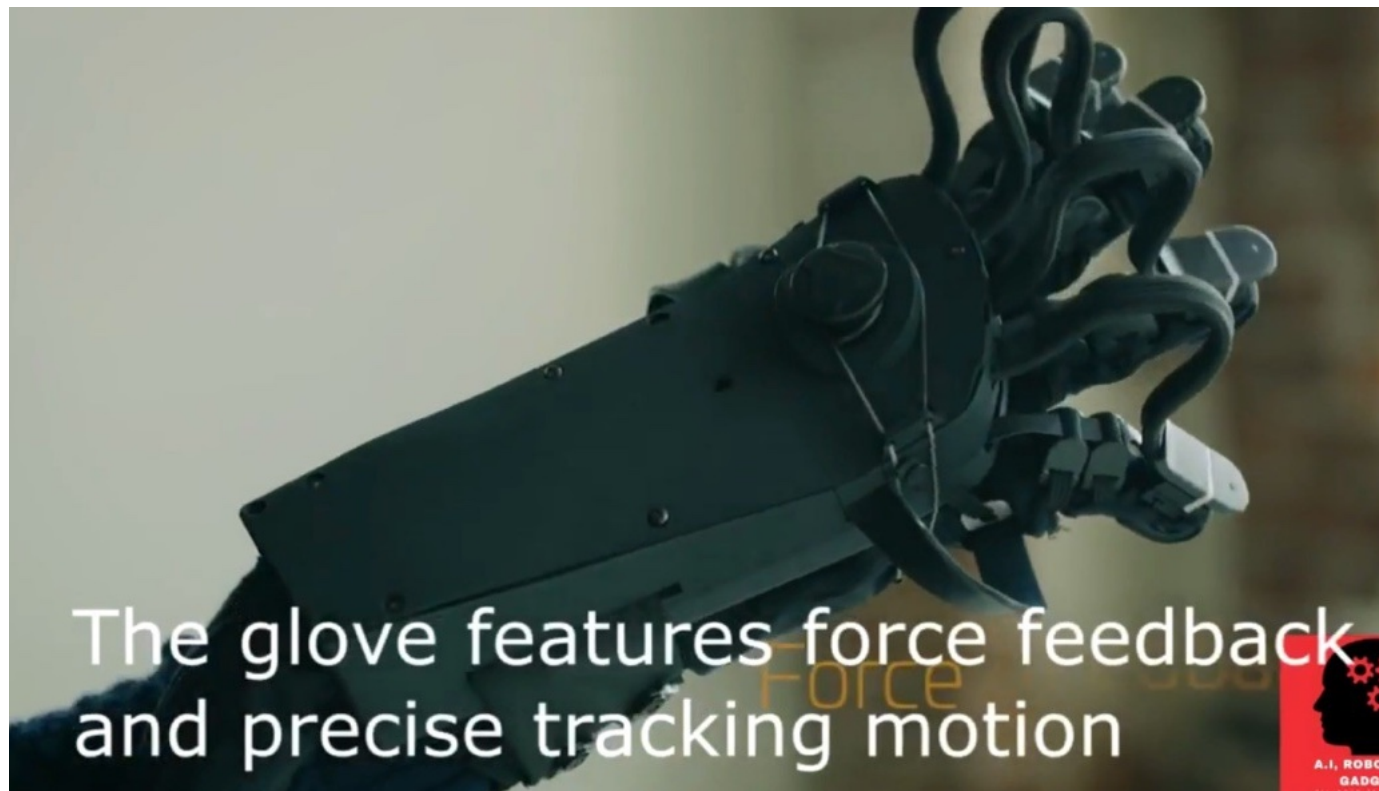
- **Goal: Stimulate skin tactile receptors**
- **Using different technologies**
 - Air bellows
 - Jets
 - Actuators (commercial)
 - Micropin arrays
 - Electrical (research)
 - Neuromuscular stimulations (research)

Vibrotactile Feedback Projects



TactaBoard and TactaVest
Navy TSAS Project

Example: HaptX Glove



https://www.youtube.com/watch?v=4K-MLVqD1_A

Audio Displays

- **Spatialization vs. Localization**
- **Spatialization** is the processing of sound signals to make them emanate from a point in space
 - This is a technical topic
- **Localization** is the ability of people to identify the source position of a sound
 - This is a human topic, i.e., some people are better at it than others.

Audio Display Properties

- Presentation Properties
 - Number of channels
 - Sound stage
 - Localization
 - Masking
 - Amplification

Audio Displays: Head-worn



Ear Buds



On Ear



Open Back



Over ear



Bone conduction

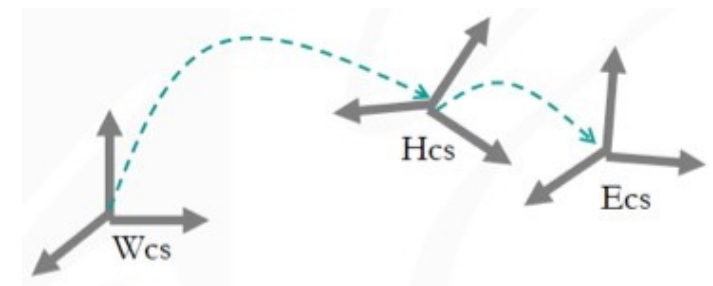
Tracking in Virtual Reality

Registration

- Positioning virtual object wrt real/virtual objects

Tracking

- Continually locating the users viewpoint/ input
- Position (x,y,z), Orientation (roll,pitch,yaw)



Mechanical Tracker

Idea: mechanical arms with joint sensors

Pros:

- High Accuracy
- Haptic Feedback

Cons:

- Cumbersome
- Restricted
- Expensive



Sutherland



Microscribe

Magnetic Tracker

Idea: Sensors calculate position by measuring induced magnetic fields

- Uses stationary base stations that generate perpendicular magnetic fields
- When an object moves through the magnetic field, its internal coils present small electrical voltages (depending on the position and orientation)

Pros:

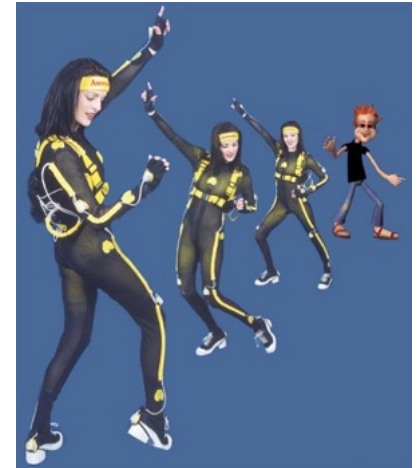
- 6-DOF, robust

Cons:

- Wired, sensible to metal, noisy, expensive
- Error increases with distance



Flock of Birds (Ascension)



Example: Razer Hydra

- Developed by Sixense
- Magnetic source + 2 wired controllers
- Short range (1-2 m)
- Precision of 1mm and 1°
- \$600 USD



Razer Hydra Demo



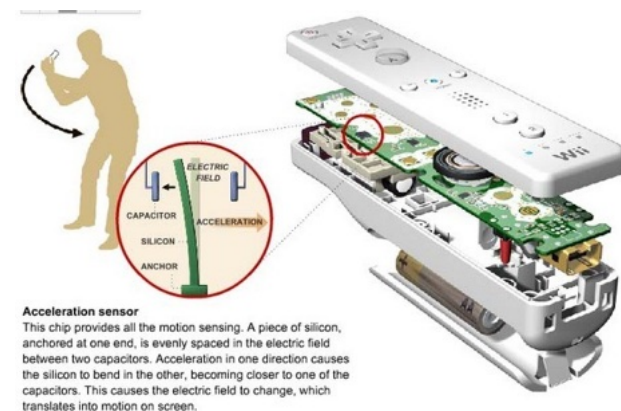
<https://www.youtube.com/watch?v=jnqFdSa5p7w>

Inertial Tracker

- **Idea:** measuring linear and angular orientation rates (accelerometer/gyroscope)
- Pros:
 - No transmitter, cheap, small, high frequency, wireless
- Cons:
 - Drift, hysteresis, only 3-DOF



IS300 (Intersense)



Wii Remote

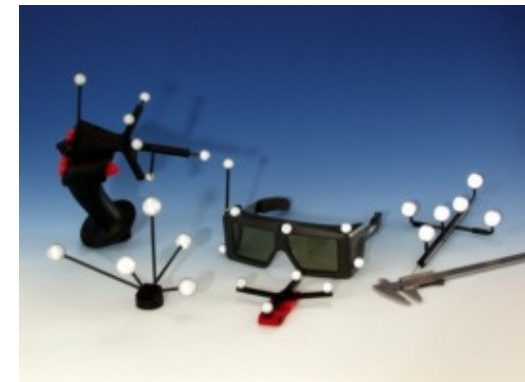
Optical Tracker

- Idea: Image Processing and Computer Vision
- Specialized: Infrared, Retro-Reflective, Stereoscopic

Monocular Based Vision Tracking

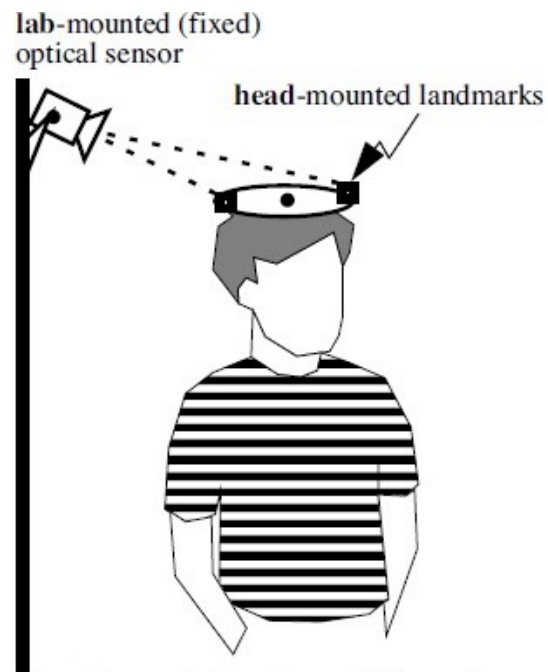


ART



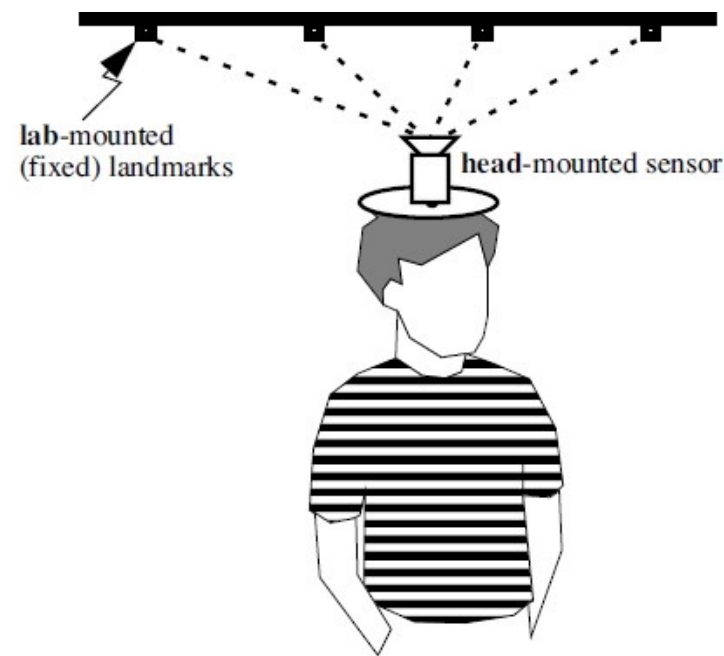
Hi-Ball

Outside-In vs. Inside-Out Tracking



Outside-Looking-In

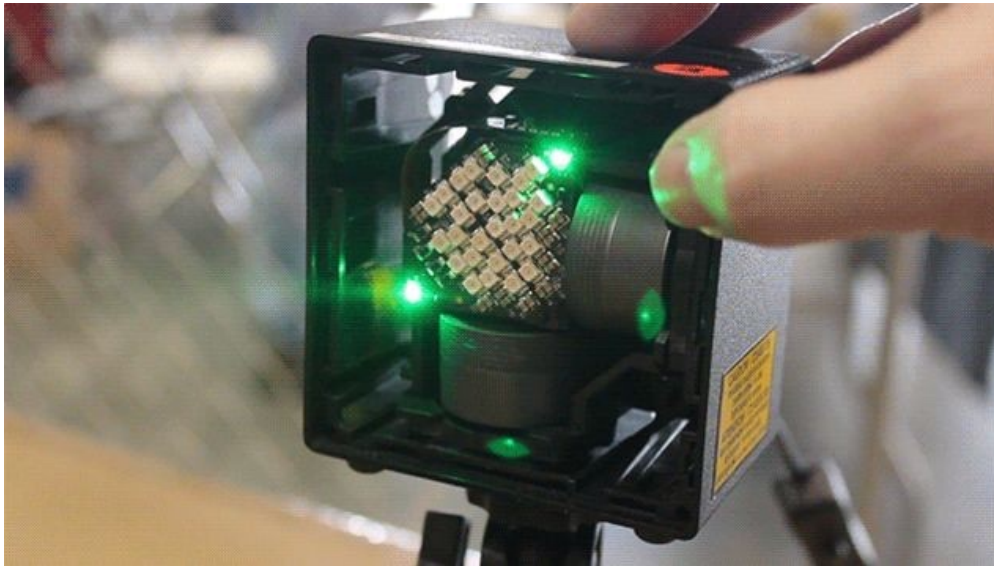
HTC Vive



Inside-Looking-Out

In the Microsoft Mixed Reality headsets the landmarks are image features

Vive Lighthouse Components

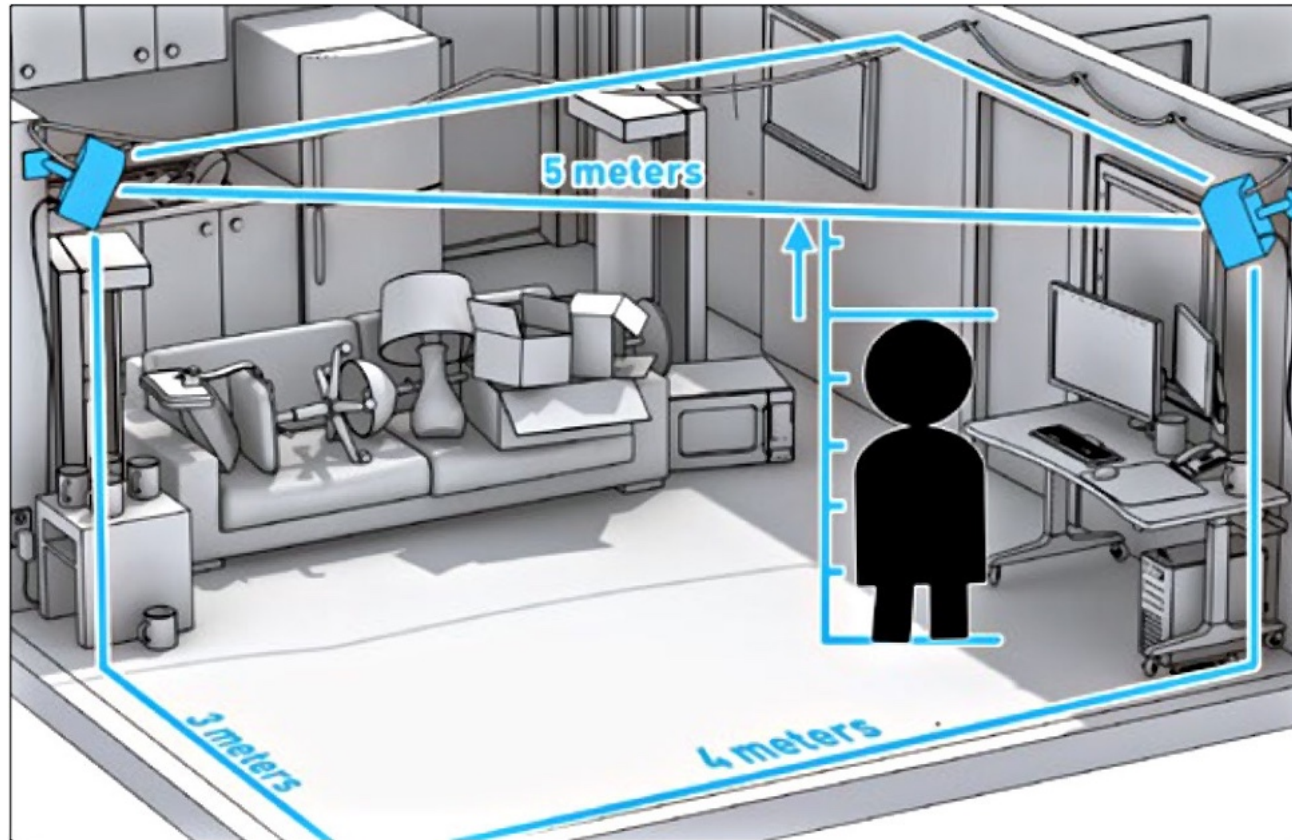


Base station
IR LED array
2 x scanned lasers
Sampling rate 250 Hz, 4 ms latency

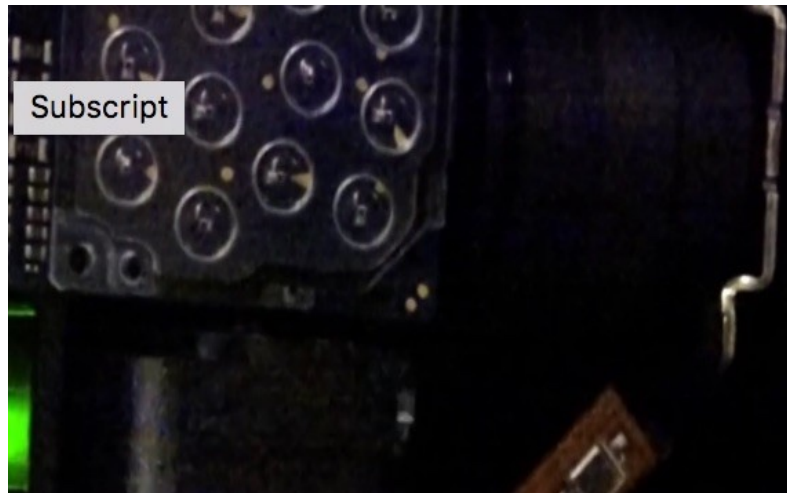


Head Mounted Display
- 37 photo sensors
- 9 axis IMU

Lighthouse Setup



Lighthouse Tracking



Base station scanning
https://www.youtube.com/watch?v=avBt_P0wg_Y



Room Tracking
<https://www.youtube.com/watch?v=oqPaaMR4kY4>

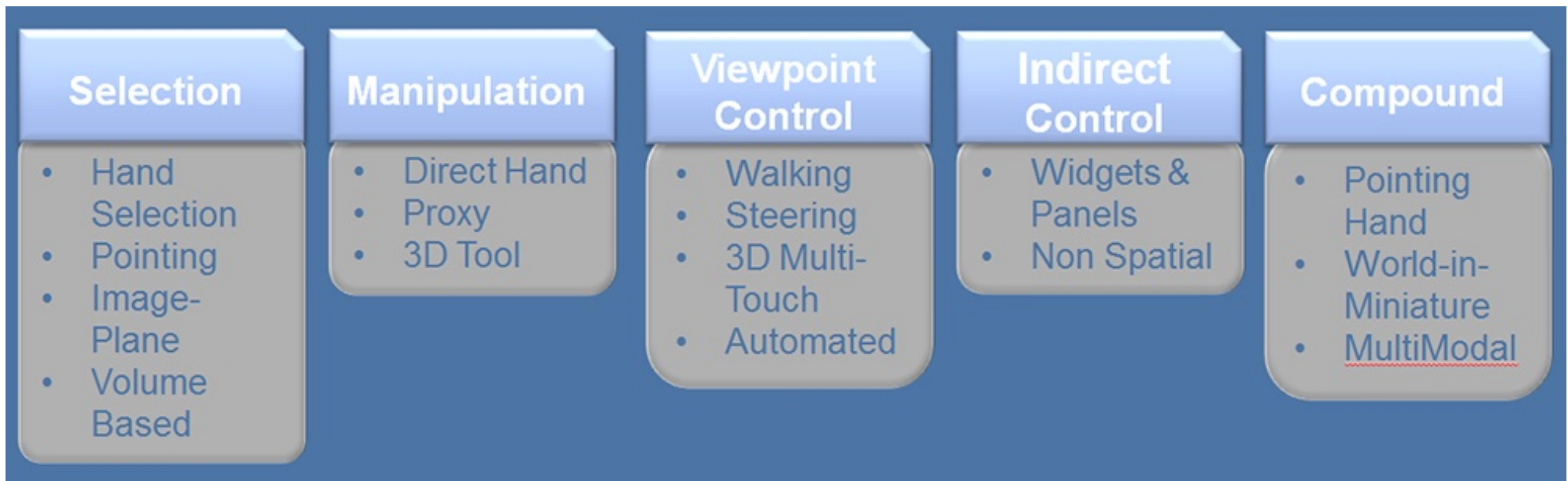
Example: Windows Mixed Reality

- Inside-out tracking
- Two cameras on HMD
- Low rez Black and White
- Combined with IMU data
- Uses environment mapping
- Tracking from visual features



VR Interaction

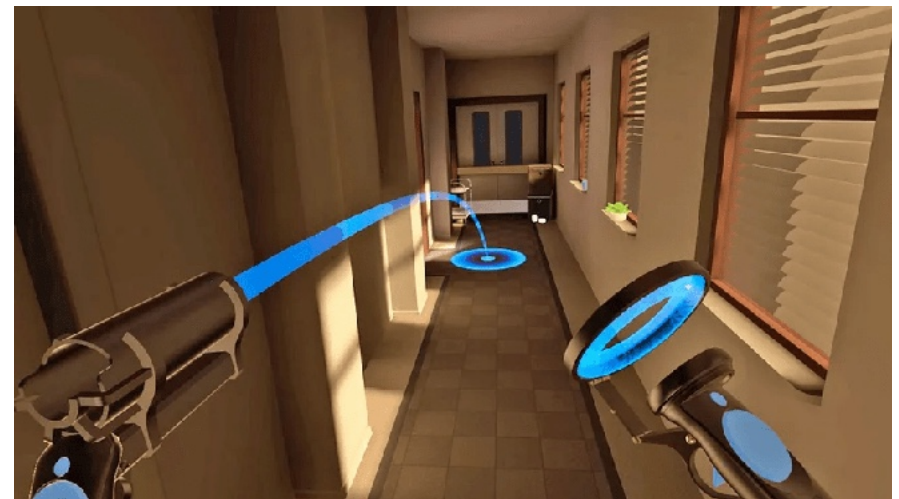
Interaction Patterns



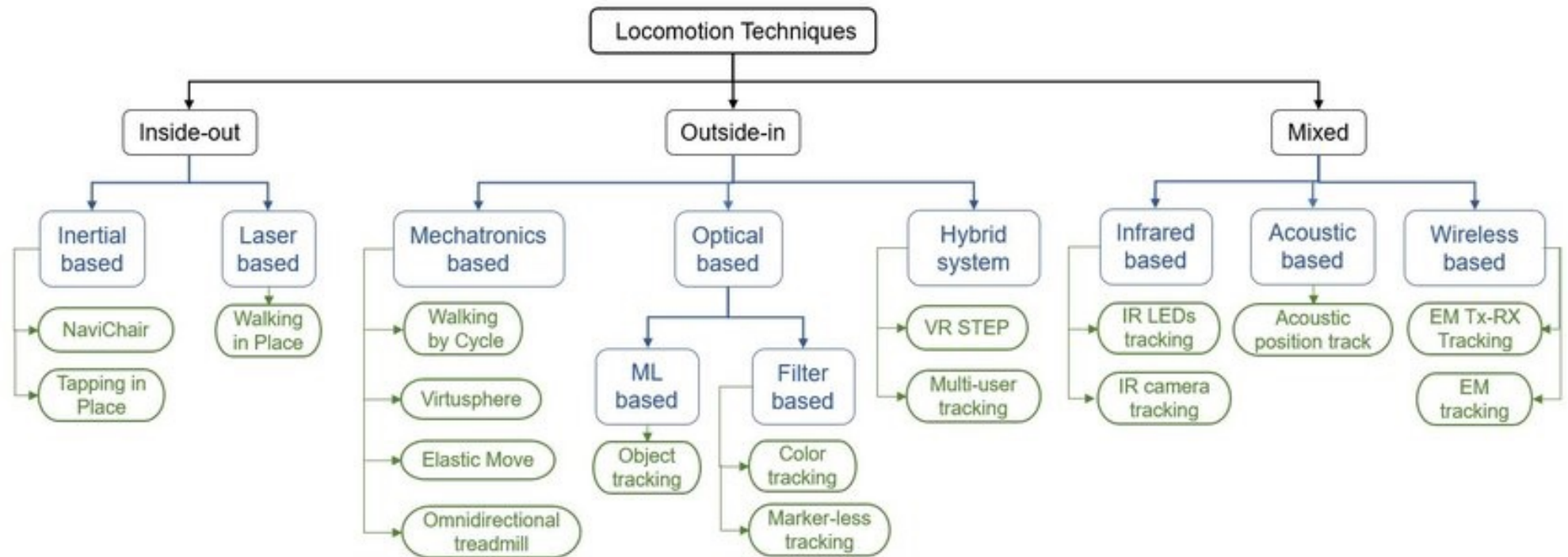
[The VR Book: Human-Centered Design for Virtual Reality](#)

Locomotion Techniques

- How to move from A to B in Virtual Reality?
- Real Space has limitations, if we keep walking we will hit a wall!!
- Main technique: Teleportation

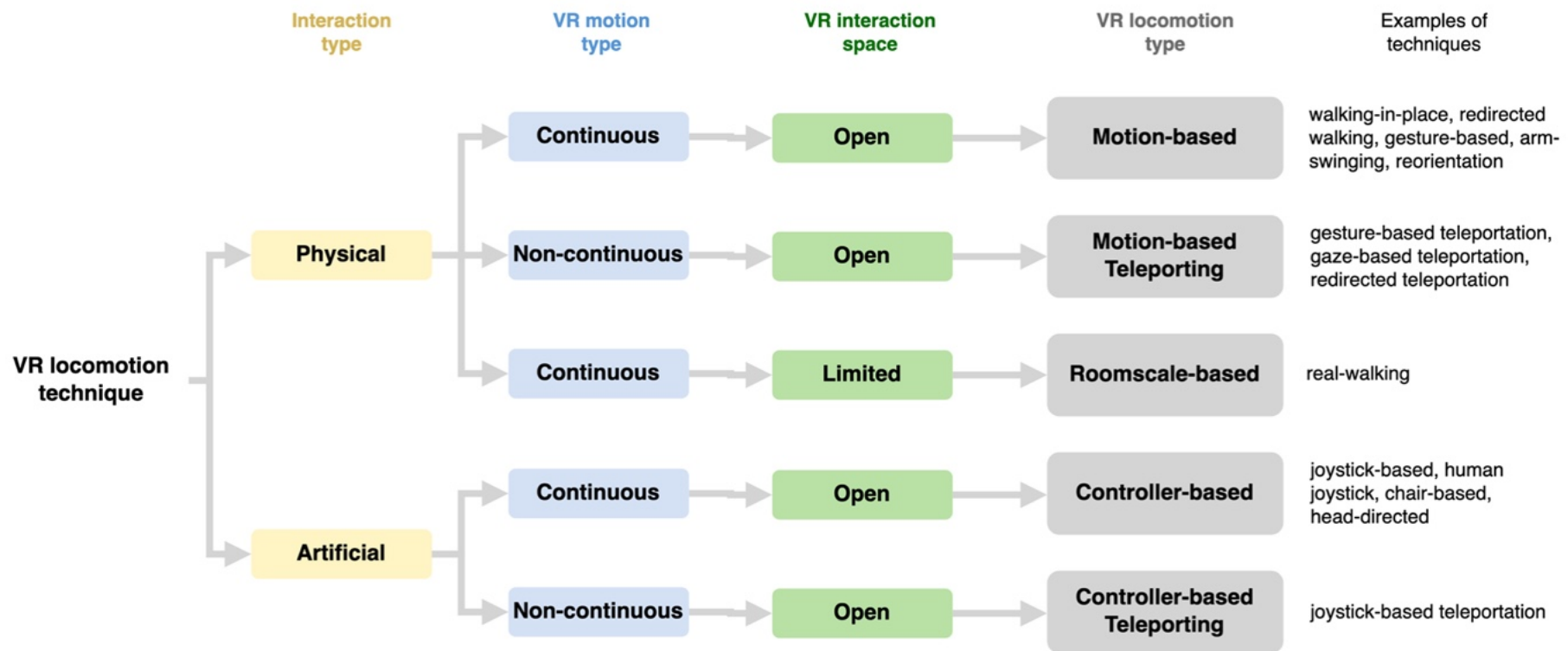


Locomotion Techniques



Gururaj, Y & Mittal, Raghav & Karre, Sai & Reddy, Raghu & Syed, Azeemuddin. (2023). Towards Conducting Effective Locomotion Through Hardware Transformation in Head-Mounted-Device -- A Review Study. 10.48550/arXiv.2306.14210.

Locomotion Techniques



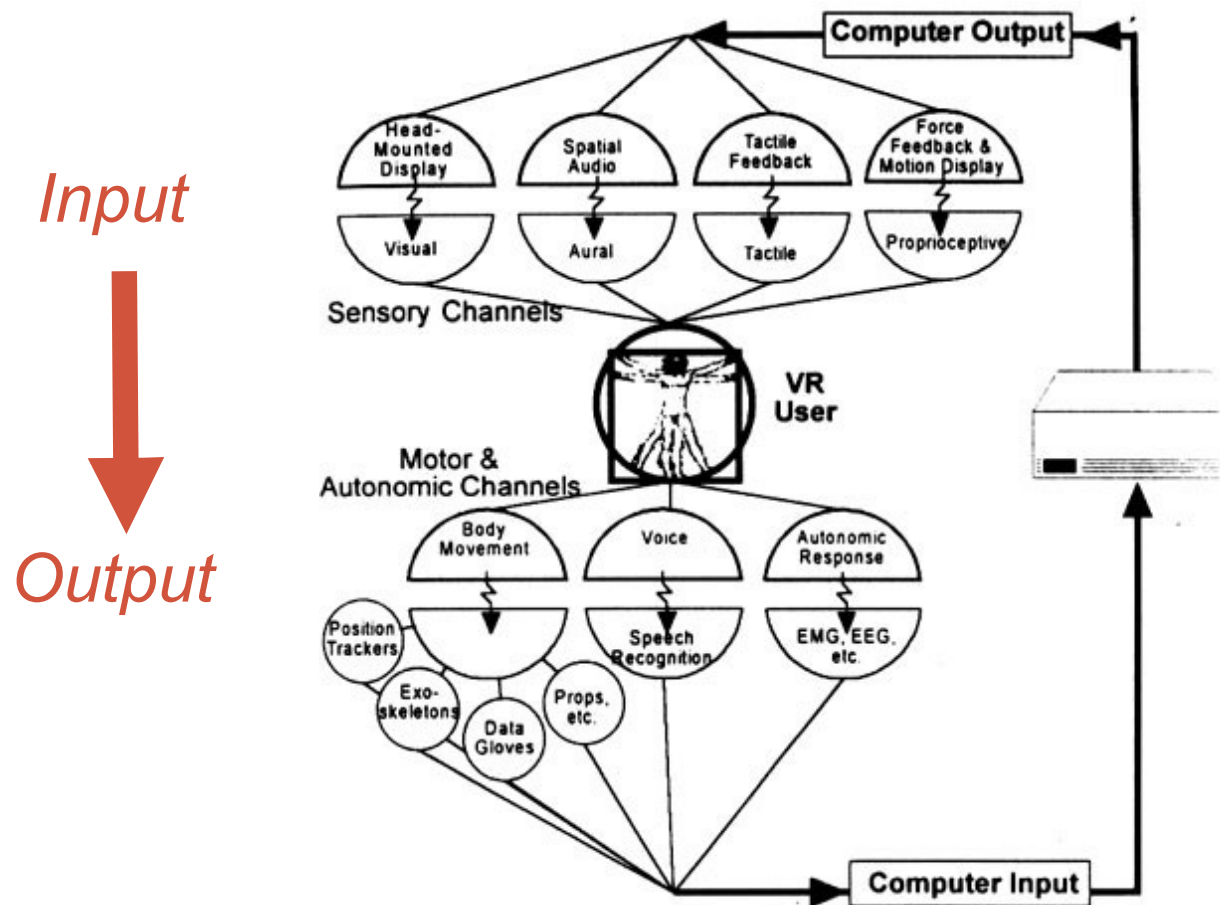
Boletsis, Costas. (2017). The New Era of Virtual Reality Locomotion: A Systematic Literature Review of Techniques and a Proposed Typology. Multimodal Technologies and Interaction. 1. 24. 10.3390/mti1040024.

VR Input Devices

Physical devices that convey information into the application and support interaction in the Virtual Environment



Mapping Between Input and Output



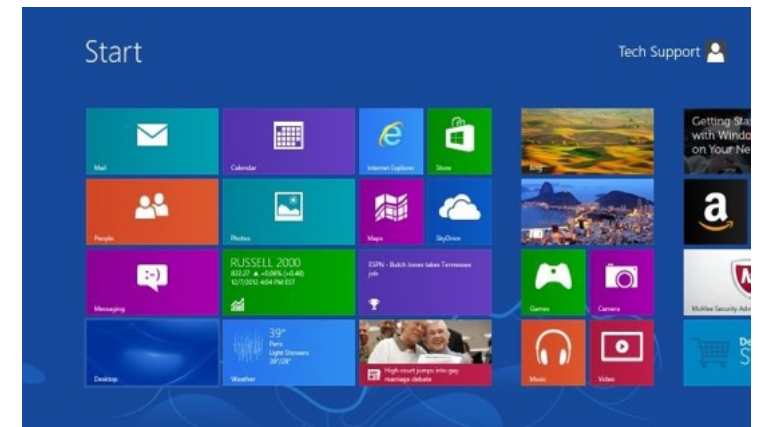
Motivation

Mouse and keyboard are good for desktop UI tasks

- Text entry, selection, drag and drop, scrolling, rubber banding, ...
- 2D mouse for 2D windows

What devices are best for 3D input in VR?

- Use multiple 2D input devices?
- Use new types of devices?

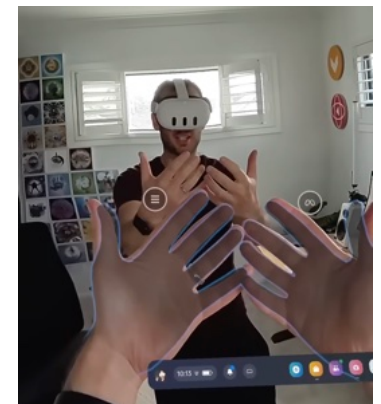
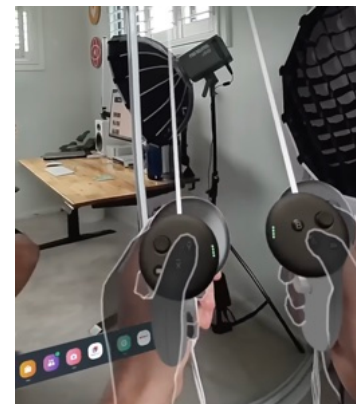
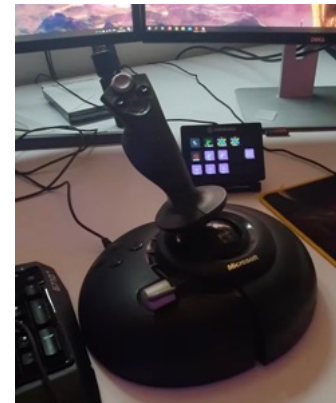


Input Device Characteristics

- **Size and shape, encumbrance**
- **Degrees of Freedom**
 - Integrated (mouse) vs. separable (Etch-a-sketch)
- **Direct vs. indirect manipulation**
- **Relative vs. Absolute input**
 - Relative: measure difference between current and last input (mouse)
 - Absolute: measure input relative to a constant point of reference (tablet)
- **Rate control vs. position control**
- **Isometric vs. Isotonic**
 - Isometric: measure pressure or force with no actual movement
 - Isotonic: measure deflection from a center point (e.g. joystick)

Hand Input Devices

- **Devices that integrate hand input into VR**
- **World-Grounded input devices**
 - Devices fixed in real world (e.g. joystick)
- **Non-Tracked handheld controllers**
 - Devices held in hand, but not tracked in 3D (e.g. xbox controller)
- **Tracked handheld controllers**
 - Physical device with 6 DOF tracking inside (e.g. Vive controllers)
- **Hand-Worn Devices**
 - Gloves, EMG bands, rings, or devices worn on hand/arm
- **Bare Hand Input**
 - Using technology to recognize natural hand input



World Grounded Devices

- **Devices constrained or fixed in real world**
- **Not ideal for VR**
- Constrains user motion
- **Good for VR vehicle metaphor**
- Used in location based entertainment (e.g. Disney Aladdin ride)



*Disney Aladdin
Magic Carpet
VR Ride*



Non-Tracker Handheld Controllers

- Devices held in hand
 - Buttons, joysticks, game controllers, etc.
- Traditional video game controllers
 - Xbox controller



Tracked Handheld Controllers

- **Handheld controller with 6 DOF tracking**
 - Combines button/joystick input plus tracking
- **One of the best options for VR applications**
 - Physical prop enhancing VR presence
 - Providing proprioceptive, passive haptic touch cues
 - Direct mapping to real hand motion

Oculus Touch
Controllers

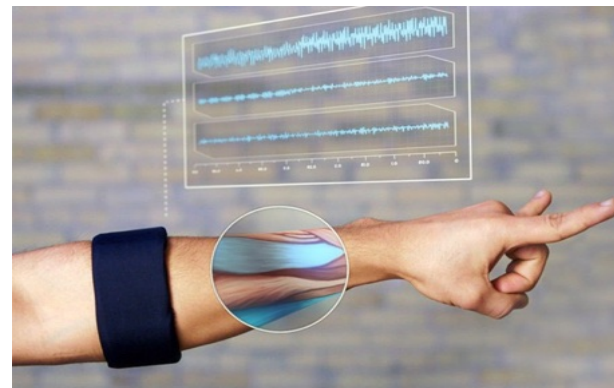


HTC Vive
Controllers



Hand Worn Devices

- **Devices worn on hands/arms**
 - Glove, EMG sensors, rings, etc.
- **Advantages**
 - Natural input with potentially rich gesture interaction
 - Hands can be held in comfortable positions – no line of sight issues
 - Hands and fingers can fully interact with real objects



Myo Arm Band



<https://www.youtube.com/watch?v=A8lGstYAY14>

Data Gloves

Bend sensing gloves

- Passive input device
- Detecting hand posture and gestures
- Continuous raw data from bend sensors
- Fiber optic, resistive ink, strain-gauge
- Large DOF output, natural hand output



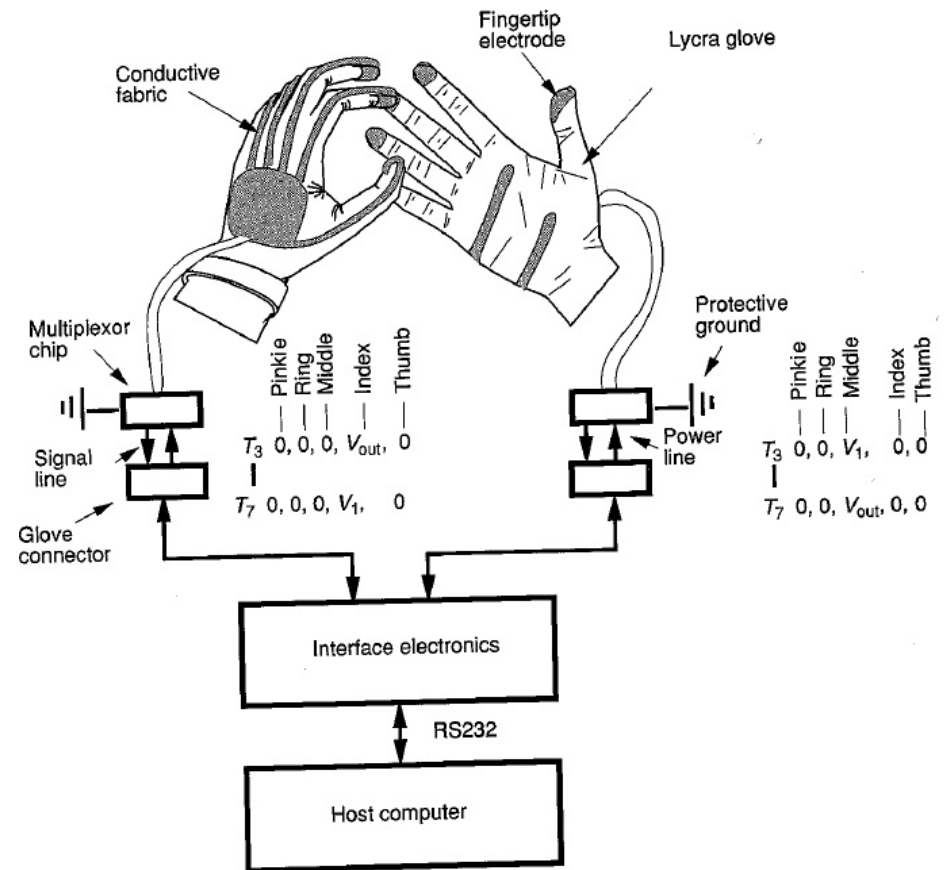
Pinch gloves

- Conductive material at fingertips
- Determine if fingertips touching
- Used for discrete input
- Object selection, mode switching, etc.



How Pinch Gloves Work

- Contact between conductive fabric completes circuit
- Each finger receives voltage in turn (T3 – T7)
- Look for output voltage at different times



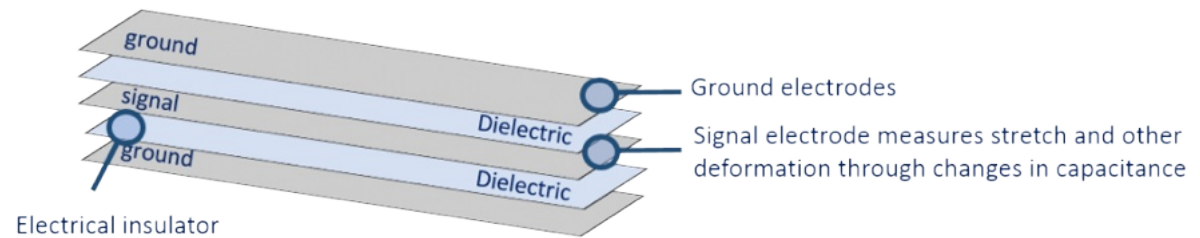
Demo Video



<https://www.youtube.com/watch?v=IUNx4FgQmas>

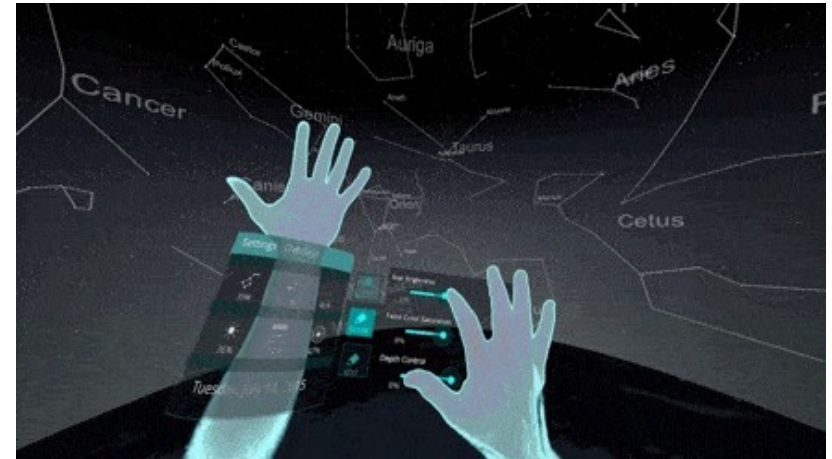
StretchSense

- **Wearable motion capture sensors**
 - Capacitive sensors
 - Measure stretch, pressure, bend, shear
- **Many applications**
 - Garments, gloves, etc.
- <http://stretchsense.com/>



Bare Hands

- **Using computer vision to track bare hand input**
- **Creates compelling sense of Presence, natural interaction**
- **Challenges needed to be solved:**
 - Not having sense of touch
 - Line of sight required to sensor
 - Fatigue from holding hands in front of sensor

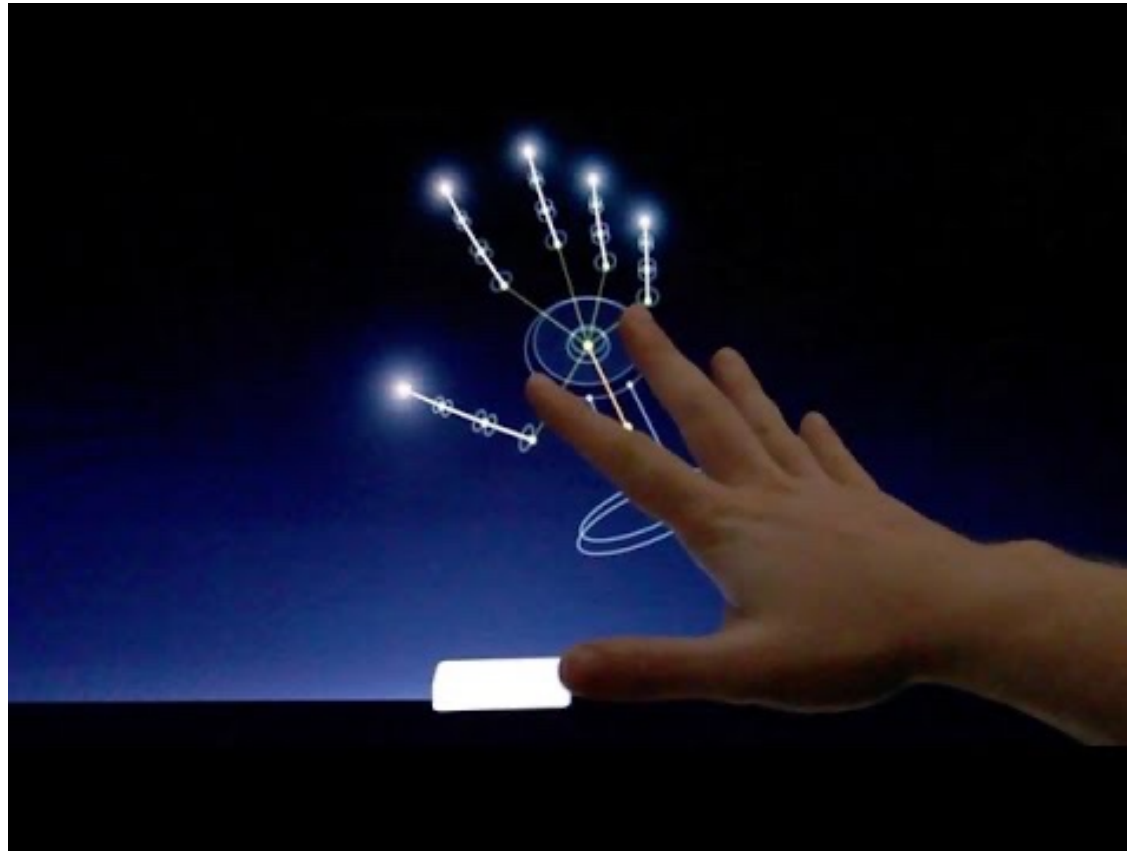


Leap Motion

- **IR based sensor for optical hand tracking (\$50 USD)**
 - HMD + Leap Motion = Hand input in VR
- **Technology**
 - 3 IR LEDS and 2 wide angle cameras
 - The LEDS generate patternless IR light
 - IR reflections picked up by cameras
 - Software performs hand tracking
- **Performance**
 - 1m range, 0.7 mm accuracy, 200Hz



Example: Leap Motion



<https://www.youtube.com/watch?v=QD4qQBL0X80>

Non-Hand Input Devices

- Capturing input from other parts of the body
- Head Tracking
 - Use head motion for input
- Eye Tracking
 - Largely unexplored for VR
- Microphones
 - Audio input, speech
- Full-Body tracking
 - Motion capture, body movement

Eye Tracking

Technology

- Shine IR light into eye and look for reflections

Advantages:

- Provides natural hands-free input
- Gaze provides cues as to user attention
- Can be combined with other input technologies



Example: FOVEE VR Headset

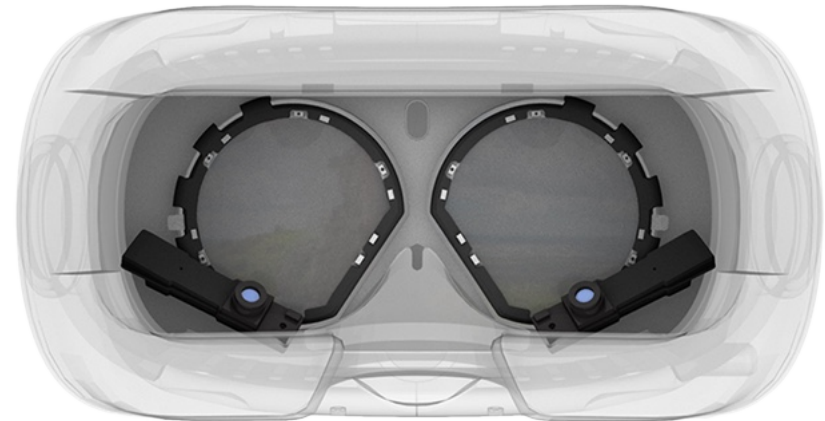
- Eye tracker integrated into VR HMD
- Gaze driven user interface, foveated rendering

<https://www.youtube.com/watch?v=cjoluz1sLK4>



Pupil Labs VIVE/Oculus Add-ons

- **Adds eye-tracking to HTC Vive/Oculus Rift HMDs**
 - Mono or stereo eye-tracking
 - 120 Hz eye tracking, gaze accuracy of 0.6° with precision of 0.08°
 - Open source software for eye-tracking
- <https://pupil-labs.com/pupil/>
- 2023: Meta Quest Pro has eye tracking



Full Body Tracking

- **Adding full-body input into VR**
 - Creates illusion of self-embodiment
 - Significantly enhances sense of Presence
- **Technologies**
 - Motion capture suit, camera based systems
 - Can track large number of significant feature points



Camera Based Motion Tracking

- Use multiple cameras
- Reflective markers on body
- Eg – Optitrack (www.optitrack.com)
- 120 – 360 fps, < 10ms latency, < 1mm accuracy



Optitrack Demo

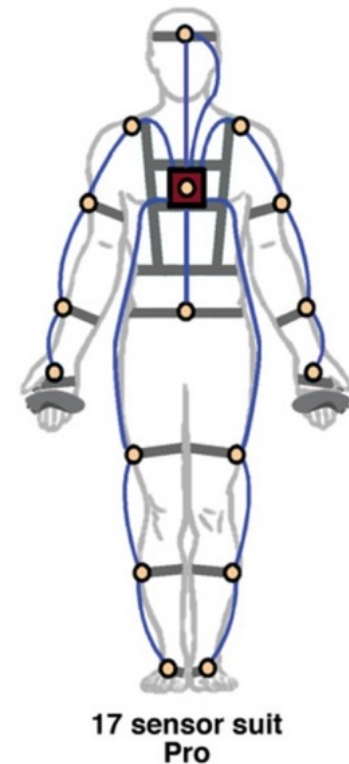
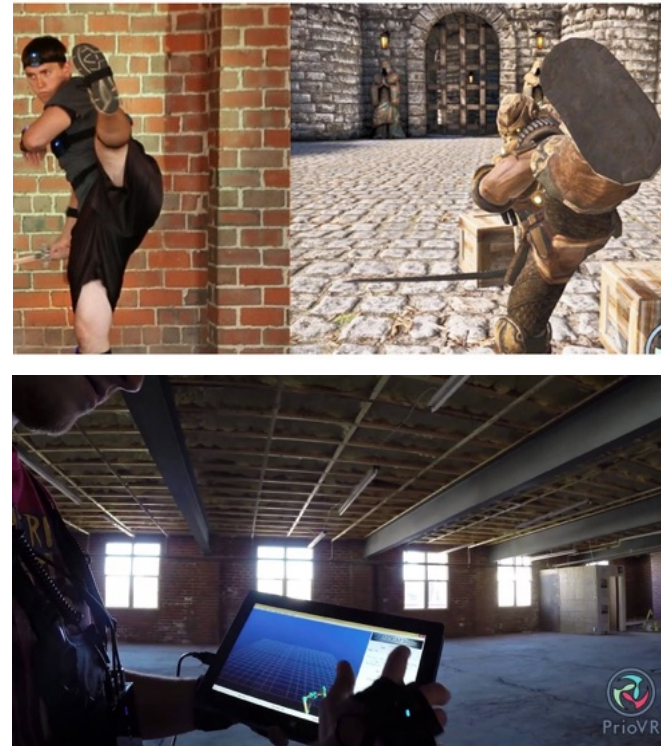


<https://www.youtube.com/watch?v=tBAvjU0Scul>

Wearable Motion Capture

- Wearable motion capture system
 - 8 – 17 inertial sensors + wireless data transmission
 - 30 – 40m range, 7.5 ms latency, 0.09% precision
 - Supports full range of motion, no occlusion

<https://www.youtube.com/watch?v=JON25SgVVFg>

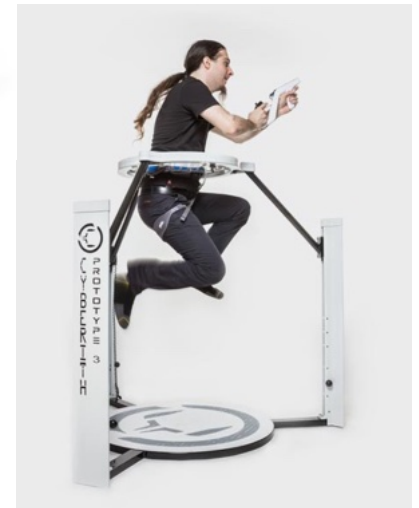


PrioVR Demo



Pedestrian Devices

- **Pedestrian input in VR**
 - Walking/running in VR
- **Virtuix Omni**
 - Special shoes
 - <http://www.virtuix.com>
- **Cyberith Virtualizer**
 - Socks + slippery surface
 - <http://cyberith.com>



Cyberith Virtualizer Demo



<https://www.youtube.com/watch?v=R8lmf3OFrms>

Virtusphere

- **Fully immersive sphere**
 - Support walking, running in VR
- Person inside trackball
- <http://www.virtusphere.com>



Virtusphere Demo



<https://www.youtube.com/watch?v=5PSFCnrk0GI>

Omnidirectional Threadmills

- Infinadeck
 - 2 axis treadmill, flexible material
 - Tracks user to keep them in centre
 - Limitless walking input in VR
- www.infinadeck.com



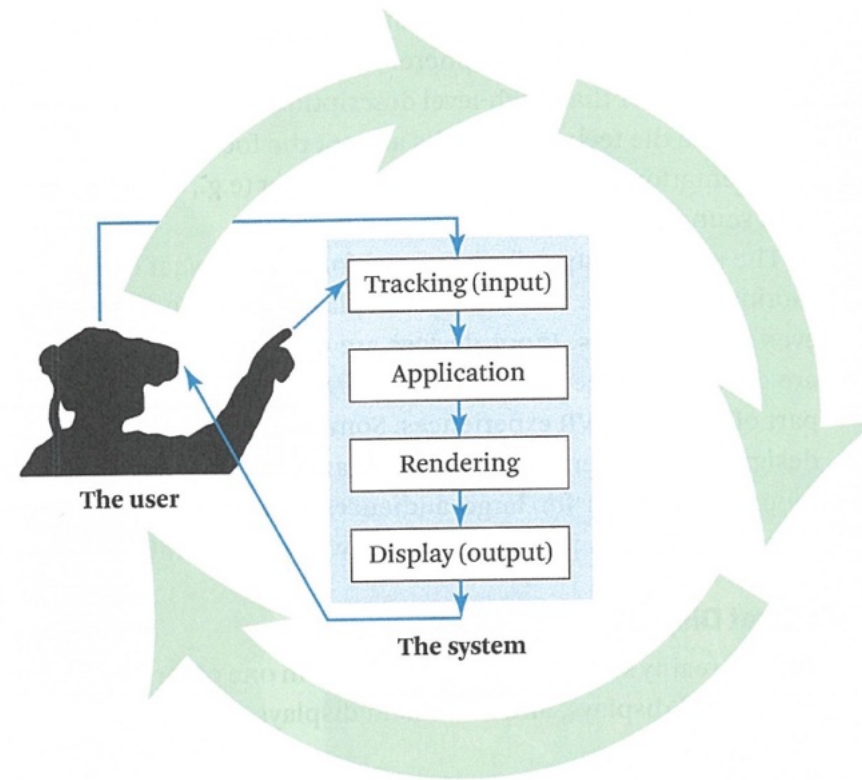
Infinadeck Demo



<https://www.youtube.com/watch?v=seML5CQBzP8>

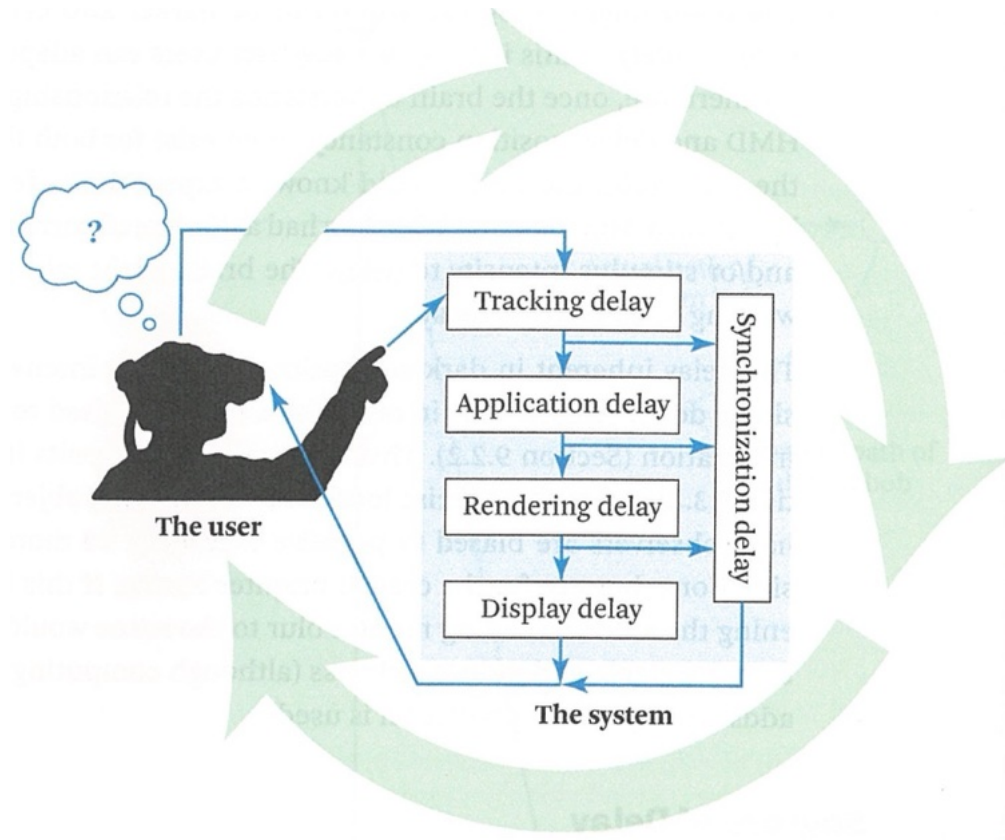
Health Issues

Typical VR Simulation Loop



User moves head, scene updates, displayed graphics change

System Delays



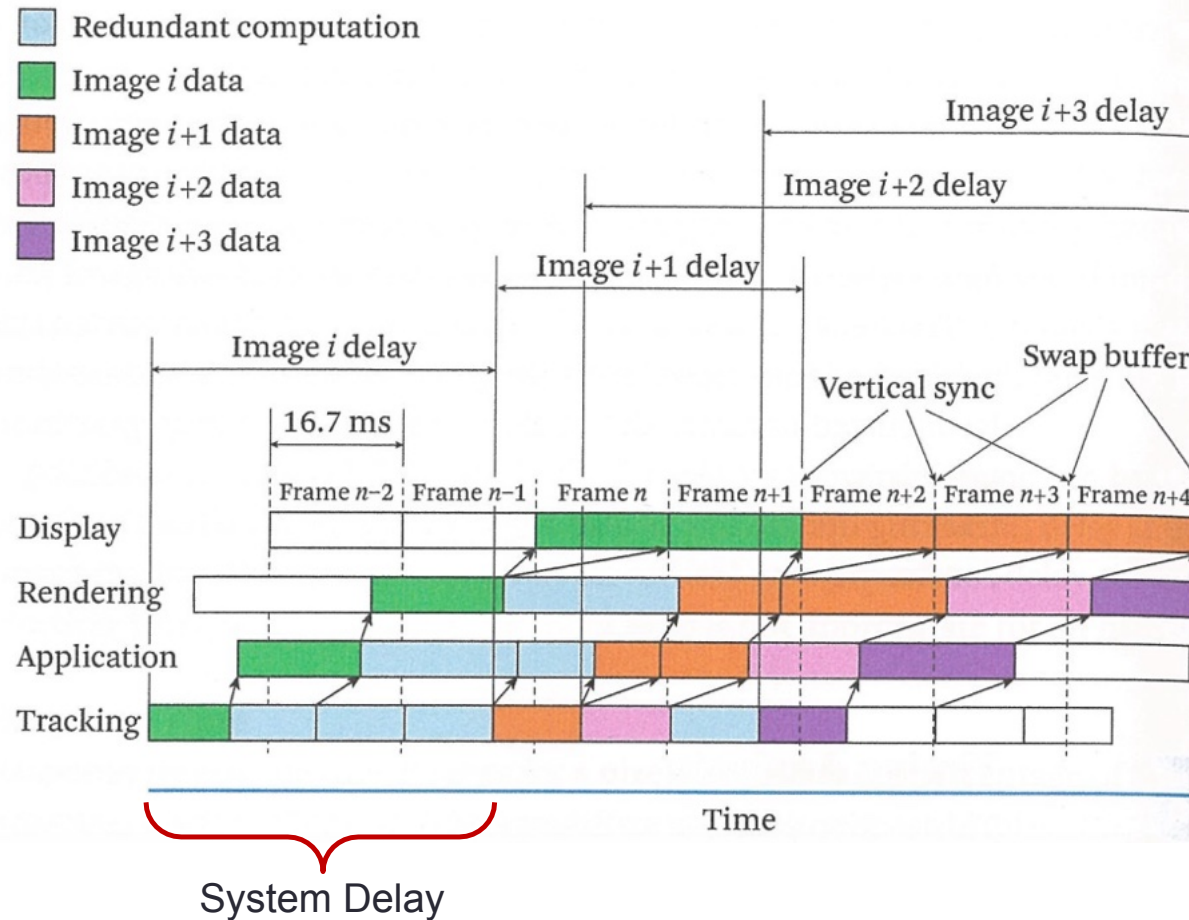
Need to synchronize system to reduce delays

Living with High Latency (1/3 sec – 3 sec)



<https://www.youtube.com/watch?v=fNp37zFn9Q>

Back to Latency



Effects of System Latency

- **Degraded Visual Acuity**

- Scene still moving when head stops = motion blur

- **Degraded Performance**

- As latency increases it's difficult to select objects etc.
- If latency > 120 ms, training doesn't improve performance

- **Breaks-in-Presence**

- If system delay high user doesn't believe they are in VR

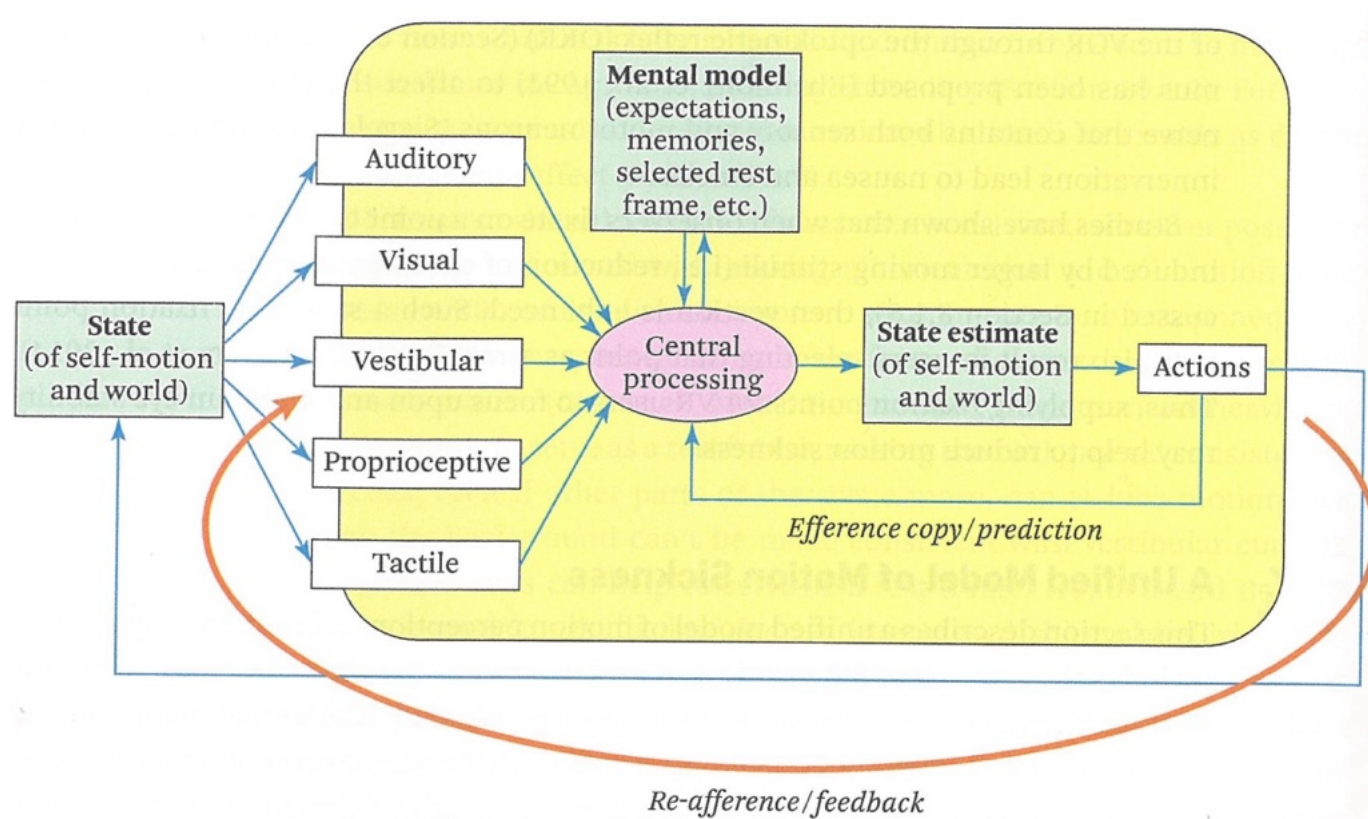
- **Negative Training Effects**

- User train to operative in world with delay

- **Simulator Sickness**

- Latency is greatest cause of simulator sickness

Simulator Sickness



Visual input conflicting with vestibular system

Many Causes of Simulator Sickness

25-40% of VR users get Simulator Sickness, due to:

- Latency (Major cause of simulator sickness)
- Tracking accuracy/precision
 - Seeing world from incorrect position, viewpoint drift
- Field of View
 - Wide field of view creates more periphery vection = sickness
- Refresh Rate/Flicker
 - Flicker/low refresh rate creates eye fatigue
- Vergence/Accommodation Conflict
 - Creates eye strain over time
- Eye separation
 - If IPD not matching to inter-image distance then discomfort

Motion Sickness



<https://www.youtube.com/watch?v=BznblIW8iqE>

CyberSickness Questionnaires

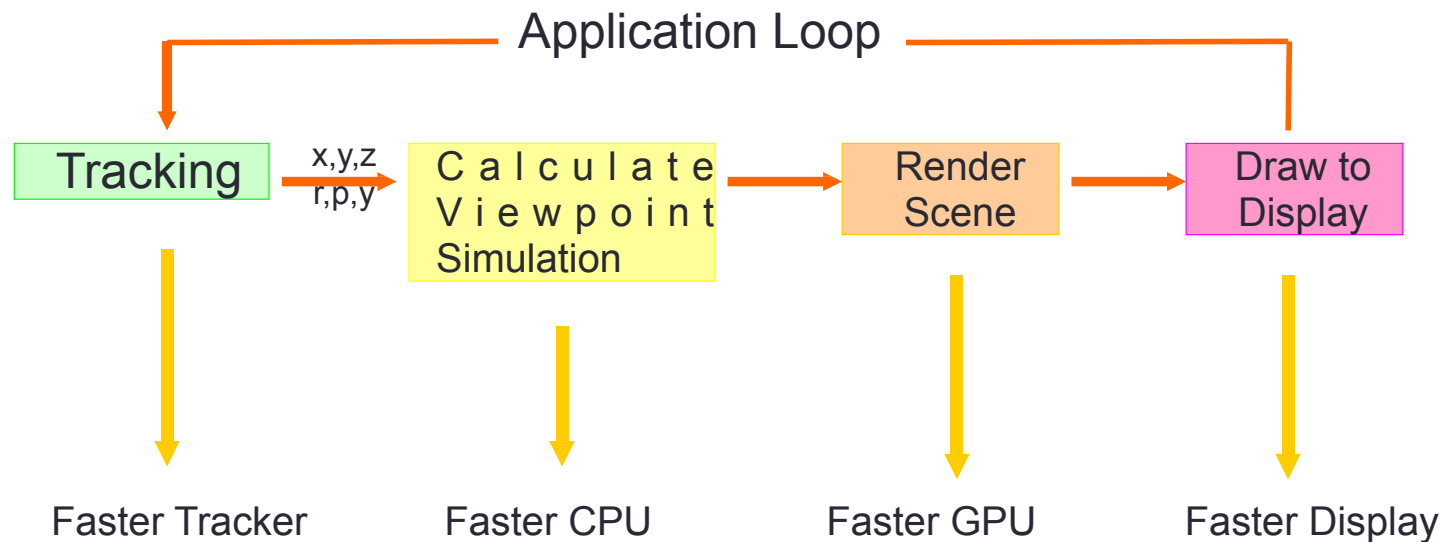
- How to evaluate cyber sickness?
- Using questionnaires
- **Simulator Sickness Questionnaire (SSQ)**

How to Reduce System Delays

- **Use faster components**
 - Faster CPU, display, etc.
- **Reduce the apparent lag**
 - Take tracking measurement just before rendering
 - Remove tracker from the loop
- **Use predictive tracking**
 - Use fast inertial sensors to predict where user will be looking
 - Difficult due to erratic head movements

Jerald, J. (2004). Latency compensation for head-mounted virtual reality. UNC Computer Science Technical Report.

Reducing System Lag



Reducing Apparent Lag

